161 D VirtuLectra

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Chapter 1

Introduction

The purpose of this project proposal document is to collect, analyze, and define high-level needs and features of our proposed app, "VirtuLectra".
This web-based application is geared towards teachers who want to lessen their burden and students who would benefit greatly from interactive lectures.

1.1. Problem Statement

Current online tutoring platforms often fall short of providing a truly interactive learning experience. Teachers face the challenge of managing diverse student needs, creating customized content, and maintaining student engagement, all while managing their existing workload. Due to this, students struggle with the lack of personalized feedback, and real-time interaction.

The problem of	Lack of interactive and efficient teaching tools for teachers, leading to increased workload and lack of student engagement.
Affects	●Teachers ● Students
The result of which	**Teachers**: ● Cannot effectively respond to students in real time, causing disengagement. ● Lack of standardized tools to automate content generation, increasing preparation time. **Students**: ● Receive limited real-time feedback, which hampers their ability to stay engaged and improve on their learning path.
Benefits of	● **Automated Content Creation**: Teachers benefit from AI-driven lecture notes and tailored quizzes, saving time on preparation and grading. ● **Real-Time Interaction**: Students enjoy dynamic interaction, ensuring real-time feedback and engagement during live sessions.

1.2. Scope

The scope of **VirtuLectra** focuses on delivering a comprehensive AI-powered educational platform designed to enhance both teaching and learning experiences. The platform supports real-time interaction between the system and students, automated content generation, and student monitoring features. By leveraging artificial intelligence, VirtuLectra aims to lessen the heavy workload on teachers by automating repetitive tasks like content creation and grading. At the same time, it enhances student engagement by enabling real-time feedback. The platform is built to adapt to diverse educational needs, offering a dynamic solution. However, the effectiveness of most of these features depends on certain limitations, such as the availability of high-quality internet connectivity.

Limitations

Dependence on Internet Connectivity: Requires stable internet for real-time interaction and content updates.

Scalability of AI Systems: Handling large numbers of students may require significant computational resources, potentially limiting performance in large-scale deployments.

Subject: Programming Fundamentals (For proof of concept)

Language: English only

Content Generation: Mcq-based quizzes only.

1.3. Modules

1.3.1. Real-Time Interaction

This module facilitates dynamic, real-time engagement between the system and students during live lectures through text-to-speech models. Key features include:

a. **Conversational Voice Exchanges:** Enables natural, interactive voice conversations between the system and students, creating a realistic and immersive classroom environment. This interaction mimics a live classroom setting.

b. **Real-time Q&A Sessions:** Utilizes LLM to address student questions when they are asked. This proactive approach helps ensure that students' queries are resolved promptly.

1.3.2. Automated Content Creation

This module streamlines the creation of educational materials using LLMs, significantly reducing the teacher's workload:

a. **Lecture Script and Notes Generation:** Automatically generates comprehensive lecture script and notes from the course content. These notes include key points, summaries, and important details.

b. **Tailored Assessments:** Creates customized quizzes aligned with the course material. These are designed to evaluate students' understanding of the content. Also automated grading of quizzes for quick feedback.

1.3.3. Student Monitoring

This module monitors and analyzes student engagement and understanding during lectures:

a. **Engagement Tracking:** Uses LLM and scheduled pop-up questions to track student attention levels and participation during live sessions. This includes monitoring student responses, and overall involvement in the lecture.

1.3.4. Web Portal

The web portal serves as the user interface for both teachers and students. It offers easy navigation and a seamless experience, allowing access to all the features of the platform:

a. **Teacher Dashboard:** Provides teachers with tools to manage lectures, review student progress, and generate content.
b. **Student Interface:** Students can access their learning materials, quizzes, and real-time lectures via the portal.
c. **Integration with Modules:** The portal integrates the Real-Time Interaction, Automated Content Creation, and Student Monitoring modules, allowing both teachers and students to interact with the AI-powered features effectively.

1.4. User Classes and Characteristics

Demographic: The platform is designed for teachers and students from a wide range of educational institutions, including schools, universities, and private tutoring programs.

User Classes:

Teachers who want to decrease their burden.

Students who want more interactive and engaging learning.

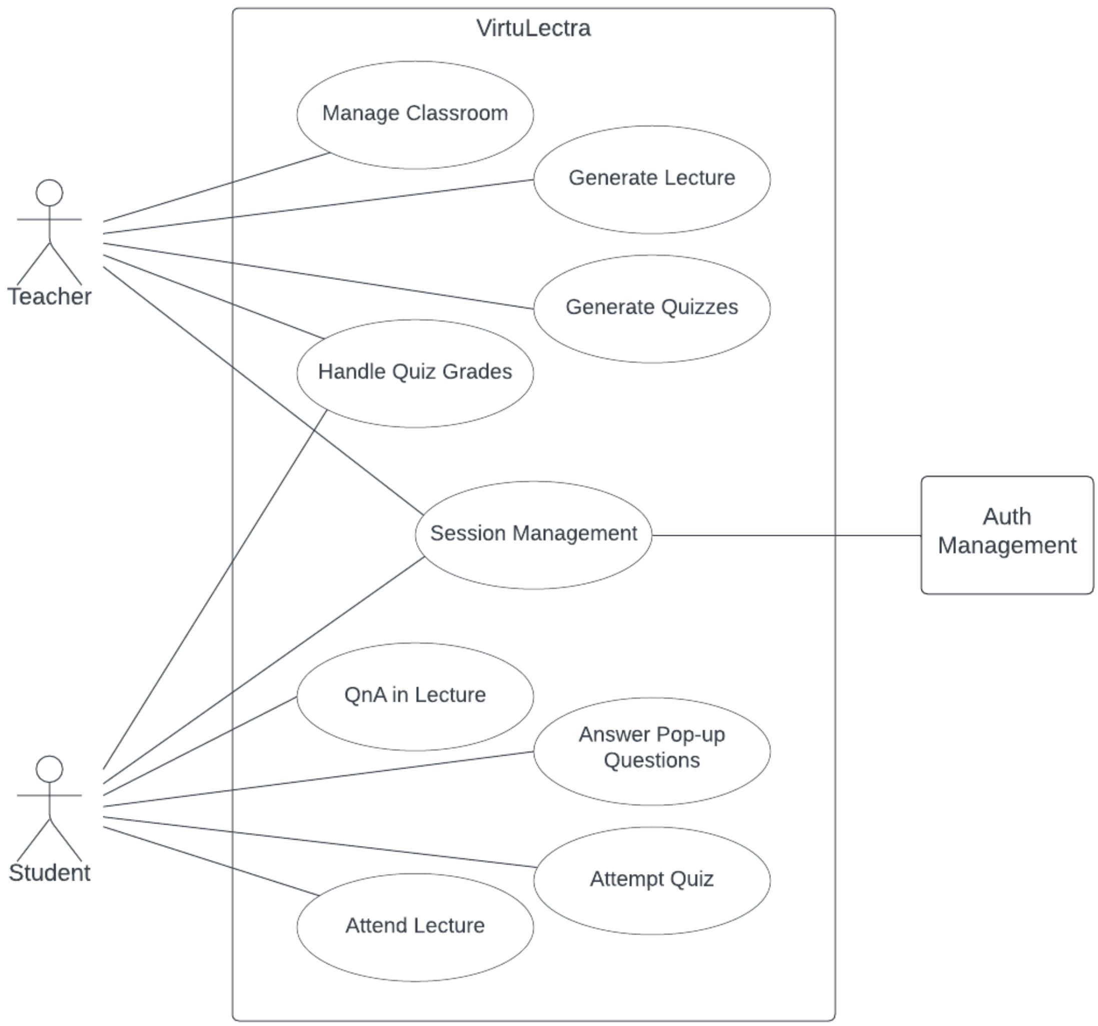
Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1. Use-case

2.1.1. Use Case Diagram



2.1.2. UC-01 Session Management

<table>
<tr><td>Use Case Name:</td><td>Session Management</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Teacher, Student</td></tr>
<tr><td>Stakeholders:</td><td>● Teacher: Needs secure and efficient access to manage classrooms, upload materials, and monitor student performance and engagement.
● Student: Requires reliable access to join classrooms, participate in lectures, and access learning materials.</td></tr>
<tr><td>Description:</td><td>Teachers and students can register using their details or OAuth providers (e.g., Google) to create accounts. Once registered, they can securely log in to access their dashboards. The system implements role-based access control, ensuring users can only access features relevant to their roles. Users can also reset forgotten passwords and log out securely.</td></tr>
<tr><td>Pre-conditions:</td><td>● The user has accessed the login/signup page.</td></tr>
<tr><td>Post-conditions</td><td>● The user is successfully authenticated and redirected to their respective dashboard.
● The system enforces role-based access control based on the user type (teacher or student).</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The user navigates to the VirtuLectra login page.</td><td></td></tr>
<tr><td>2. The user enters their username and password into the login form.</td><td></td></tr>
<tr><td>3. The user clicks the Login button to submit their credentials.</td><td></td></tr>
</table>
</td></tr>
</table>

	<table><tr><td></td><td>4. The system validates the entered credentials against the registered user database.</td></tr><tr><td></td><td>5. If the credentials are valid, the system authenticates the user.</td></tr><tr><td></td><td>6. The system redirects the user to their respective dashboard (Teacher Dashboard or Student Interface) based on their role.</td></tr><tr><td></td><td>7. The system enforces role-based access control, ensuring users can only access features pertinent to their roles.</td></tr></table>
Alternate flows:	1. **Forgot Password:** ○ **Trigger:** The user clicks on the "Forgot Password" link. ○ **User Action:** The user provides their registered email address to receive a password reset link. ○ **System Response:** Sends a password reset email and notifies the user to check their inbox. 2. **Signup:** ○ **Trigger:** The user selects the signup option. 3. **OAuth:** ○ **Trigger:** The user selects the signup option using an OAuth provider (e.g., Google). ○ **User Action:** The user authenticates via the chosen OAuth provider. ○ **System Response:** Grants access and navigates the user to their dashboard upon successful verification through the OAuth provider. 4. **Logout** ○ **Trigger:** The user decides to end their session. ○ **User Action:** The user clicks the "Logout" button on their dashboard. ○ **System Response:** Ends the user session, clears any session data, and redirects the user to the login page with a successful logout confirmation message.

Exceptions:	1. **Invalid Credentials - Login:** ○ **Trigger:** The user enters an incorrect username or password. ○ **System Response:** Displays an error message such as "Invalid username or password. Please try again.". ○ **User Action:** The user re-enters their credentials or uses the "Forgot Password" feature to reset their password. 2. **Invalid Credentials - Signup:** ○ **Trigger:** The user enters invalid information during the signup process (e.g., email already in use, weak password). ○ **System Response:** Displays an error message such as "Email is already registered" or "Password does not meet the required criteria.". ○ **User Action:** The user corrects the information (e.g., uses a different email, chooses a stronger password) and attempts to sign up again. 3. **Session Timeout:** ○ **Trigger:** The user is inactive for a predefined period. ○ **System Response:** Logs the user out automatically and displays a session timeout message. ○ **User Action:** The user must log in again to continue using the platform.

2.1.3. UC-02 Manage Classroom

<table>
<tr><td>Use Case Name:</td><td>Manage Classroom</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Teacher</td></tr>
<tr><td>Stakeholders:</td><td>Teacher: Needs to efficiently create and manage virtual classrooms, upload course materials, and monitor student performance and engagement.
Student: Benefits from organized classroom environments and accessible course materials.</td></tr>
<tr><td>Description:</td><td>Teachers can create and manage virtual classrooms by uploading course materials and configuring classroom settings. Teachers can invite students using unique codes, edit or delete classrooms, and update classroom information as needed.</td></tr>
<tr><td>Pre-conditions:</td><td>The teacher is logged into VirtuLectra.
The teacher has course materials ready in supported formats (PDF, DOCX).</td></tr>
<tr><td>Post-conditions</td><td>A new virtual classroom is created and accessible to students.
Course materials are uploaded and linked to the classroom.</td></tr>
<tr><td>Main Scenario:</td><td><table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The teacher navigates to the Create Classroom section in the portal.</td><td></td></tr>
<tr><td></td><td>2. The system displays a form prompting the teacher to enter classroom details, such as name, description, etc.</td></tr>
<tr><td>3. The teacher fills in the required details.</td><td>.</td></tr>
<tr><td></td><td>4. The system presents an upload interface for course</td></tr>
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<tr><td></td><td>materials, with the option to specify which portion(s) to focus on.</td></tr>
<tr><td>5. The teacher uploads course materials to the system.</td><td></td></tr>
<tr><td></td><td>6. The system verifies the file formats and checks for any corruption or unsupported formats.</td></tr>
<tr><td></td><td>7. Upon successful upload, the system processes the material and stores it linking it to the classroom.</td></tr>
<tr><td>8. The teacher publishes the classroom, which generates a unique classroom code for student enrollment.</td><td></td></tr>
<tr><td>9. The teacher shares the classroom code with students to invite them to join the classroom.</td><td></td></tr>
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<tr>
<td>Alternate Flows:</td>
<td>
1. Edit Classroom Details:
○ Trigger: The teacher wants to update classroom information after creation.
○ User Action: The teacher navigates to the Manage Classroom section and selects the option to Edit Classroom.
○ System Response: The system displays the current classroom details for editing.
○ User Action: The teacher modifies details such as the classroom name, and description and saves the changes.
○ System Response: The system updates the classroom information and notifies the teacher of the successful update.
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</tr>
</table>

	2. Invite Students: ○ **Trigger**: The teacher wants to add students to the classroom. ○ **User Action**: The teacher shares the unique classroom code with the students. ○ **System Response**: The system allows students to join using the provided code, updating the classroom enrollment list accordingly. **3. Delete Classroom**: ○ **Trigger**: The teacher decides to remove an existing classroom. ○ **User Action**: The teacher selects the option to delete the classroom and confirms the action. ○ **System Response**: The system permanently deletes the classroom and all associated data, notifying the teacher of the successful deletion.
Exceptions:	**1. Duplicate Classroom Name**: ○ **Trigger**: The teacher attempts to create a classroom with a name that already exists. ○ **System Response**: Displays an error message like "Classroom name already in use. Please choose a different name.". ○ **User Action**: The teacher selects a unique name and retries creating the classroom. **2. Unsupported File Format Upload**: ○ **Trigger**: The teacher uploads a course material file in an unsupported format. ○ **System Response**: Notifies the teacher with "Unsupported file format. Please upload PDF or DOCX files.". ○ **User Action**: The teacher converts the file to a supported format and uploads it again. **3. Failed Content Processing**: ○ **Trigger**: The system encounters an error while processing uploaded course materials. ○ **System Response**: Alerts the teacher with "Failed to process uploaded content. Please try again later.". ○ **User Action**: The teacher attempts to re-upload the content or contacts support for assistance.

2.1.4. UC-03 Generate Lecture

<table>
<tr><td>Use Case Name:</td><td>Generate Lecture</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Teacher</td></tr>
<tr><td>Stakeholders:</td><td>Teacher: Wants to efficiently create comprehensive lecture scripts and notes tailored to specific topics and durations.
Student: Benefits from well-structured and informative lectures that enhance understanding and learning.</td></tr>
<tr><td>Description:</td><td>Teachers can generate lectures, including scripts and corresponding notes, by specifying the lecture topic and duration. The system creates the lecture based on these inputs.</td></tr>
<tr><td>Pre-conditions:</td><td>The student is logged into VirtuLectra.
A classroom has been created and is active.
Relevant course materials have been uploaded and processed by the system.</td></tr>
<tr><td>Post-conditions</td><td>The lecture is generated and available for publishing.
Scheduled lectures are accessible to students within the classroom.</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The teacher initiates the process of generating a lecture.</td><td></td></tr>
<tr><td></td><td>2. The system prompts the teacher to enter the lecture topic, specify the duration, and upload the date and time</td></tr>
<tr><td>3. The teacher provides the lecture details.</td><td>.</td></tr>
<tr><td></td><td>4. The system generates the lecture script and corresponding notes based on the provided inputs.</td></tr>
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</td></tr>
</table>

	5. The teacher publishes the lecture, making it available to students within the classroom.
Alternate Flows:	
Exceptions:	**1. Generation Failure**: ○ **Trigger**: The system fails to generate the lecture script and notes due to insufficient content or processing errors. ○ **System Response**: Notifies the teacher with an error message such as "Unable to generate lecture. Please try again later." ○ **User Action**: The teacher can retry the generation process or contact support for assistance. **2. Generation Delay**: ○ **Trigger**: The system takes longer than expected to generate the lecture script and notes due to large inputs or server issues. ○ **System Response**: Displays a progress indicator and notifies the teacher once the generation process is complete. ○ **User Action**: The teacher waits for the notification or chooses to cancel the process if necessary. **3. Partial Upload Failure**: ○ **Trigger**: Some course materials fail to process correctly, affecting content generation. ○ **System Response**: Indicates which specific materials caused the issue and provides options to retry uploading or replace them. ○ **User Action**: The teacher retries uploading the problematic materials.

2.1.5. UC-04 Generate Quizzes

<table>
<tr><td>Use Case Name:</td><td>Generate Quizzes</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Teacher</td></tr>
<tr><td>Stakeholders:</td><td>Teacher: Wants to efficiently create, review, and manage quizzes to assess student understanding.
Student: Benefits from receiving well-structured quizzes that reinforce learning and provide feedback.</td></tr>
<tr><td>Description:</td><td>Teachers can generate quizzes based on lecture topics, preview and edit the quizzes, and manage their publication. The system automates the creation of quiz questions and allows teachers to customize and finalize quizzes before making them available to students.</td></tr>
<tr><td>Pre-conditions:</td><td>The teacher is logged into their VirtuLectra account.
A classroom has been created and is active.
Relevant course materials have been uploaded and processed by the system.</td></tr>
<tr><td>Post-conditions</td><td>The quiz is generated and available for publishing.
The finalized quiz is accessible to students with a specified deadline for attempting.</td></tr>
<tr><td>Main Scenario:</td><td><table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The teacher selects the option to generate a quiz within the classroom management interface.</td><td></td></tr>
<tr><td></td><td>2. The system prompts the teacher to specify the lecture topic, time limit, and number of questions for which the quiz should be generated.</td></tr>
<tr><td>3. The teacher inputs the required details.</td><td></td></tr>
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<tr><td></td><td>4. The system analyzes the relevant course materials and generates a set of multiple-choice questions (MCQs) aligned with the specified topic.</td></tr>
<tr><td></td><td>5. The system provides a preview of the generated quiz, including questions and answer options.</td></tr>
<tr><td>6. The teacher finalizes the quiz and publishes it, making it available to students for completion.</td><td></td></tr>
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</tr>
<tr>
<td>Alternate flows:</td>
<td>
1. Edit Quiz Content:
○ Trigger: The teacher wants to modify specific quiz questions or answer options.
○ User Action: The teacher selects a question within the preview and edits its content or the associated answer choices.
○ System Response: The system saves the changes and updates the quiz accordingly.
2. Customize Quiz Settings:
○ Trigger: The teacher wants to adjust quiz parameters such as the number of questions.
○ User Action: The teacher modifies the quiz settings before finalizing.
○ System Response: The system regenerates the quiz based on the updated settings, providing a new preview for review.
3. Preview Quiz:
○ Trigger: The teacher wants to review the entire quiz before making it available to students.
○ User Action: The teacher selects the option to preview the full quiz.
○ System Response: The system displays the complete quiz for a comprehensive review.
4. Delete Generated Quiz:
○ Trigger: The teacher decides to remove a generated quiz before publishing.
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</table>

	○ **User Action**: The teacher selects the option to delete the quiz. ○ **System Response**: The system removes the quiz from the generation queue and notifies the teacher of the successful deletion.
Exceptions:	1. **Generation Failure**: ○ **Trigger**: The system fails to generate a quiz due to processing errors. ○ **System Response**: Displays an error message such as "Unable to generate quiz at this time. Please try again later.". ○ **User Action**: The teacher can attempt to regenerate the quiz after addressing the issues or contact support for assistance. 2. **Partial Quiz Generation**: ○ **Trigger**: Some quiz questions fail to generate due to content gaps. ○ **System Response**: Informs the teacher which specific questions cannot be generated and provides options to manually create those questions. ○ **User Action**: The teacher manually adds the missing questions or adjusts the lecture topic parameters to facilitate complete quiz generation. 3. **System Timeout**: ○ **Trigger**: The system takes longer than expected to generate the quiz due to large content size or server issues. ○ **System Response**: Displays a timeout notification and offers the teacher the option to retry or cancel the quiz generation process. ○ **User Action**: The teacher can choose to retry the generation process or cancel and address any underlying issues.

2.1.6. UC-05 Attend Lecture

<table>
<tr><td>Use Case Name:</td><td>Attend Lecture</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Student</td></tr>
<tr><td>Stakeholders:</td><td>Student: Wants to navigate to and participate in lectures effectively.</td></tr>
<tr><td>Description:</td><td>Students attend scheduled lectures by accessing their classrooms and joining the lecture sessions. The system delivers lecture content based on predefined scripts and displays accompanying lecture notes, ensuring a personalized and focused learning experience.</td></tr>
<tr><td>Pre-conditions:</td><td>The student is enrolled in the classroom.
A lecture session is active.
Lecture materials have been generated and are available.</td></tr>
<tr><td>Post-conditions</td><td>The student has attended the lecture.
Lecture notes are accessible.</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The student accesses their enrolled classroom.</td><td></td></tr>
<tr><td></td><td>2. The system displays the list of scheduled lectures.</td></tr>
<tr><td>3. The student selects and joins the scheduled lecture session.</td><td></td></tr>
<tr><td></td><td>4. The system initiates the lecture, delivering content based on the generated script and displaying lecture notes in the background.</td></tr>
<tr><td></td><td>5. The system manages the flow of the lecture, ensuring content is delivered clearly.</td></tr>
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</table>

	6. The student listens to the lecture content and follows along with the notes.
Alternate Flows:	
Exceptions:	1. **Network Issues**: ○ **Trigger**: The student loses internet connectivity during the lecture. ○ **System Response**: Alerts the student of the connectivity issue and attempts to reconnect automatically. 2. **System Error During Lecture Delivery**: ○ **Trigger**: The system encounters an error while delivering the lecture content. ○ **System Response**: Notifies the student of the issue and offers options to retry joining or contact support.

2.1.7. UC-06 Q&A in Lecture

<table>
<tr><td>Use Case Name:</td><td>Q&A in Lecture</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Student</td></tr>
<tr><td>Stakeholders:</td><td>Student: Wants to ask questions during lectures to clarify doubts and enhance understanding.
Teacher: Seeks to facilitate interactive sessions and address student queries effectively.</td></tr>
<tr><td>Description:</td><td>During a lecture, students can submit questions to seek clarification or further information. The system processes these queries and provides timely responses, enhancing engagement and understanding.</td></tr>
<tr><td>Pre-conditions:</td><td>The student is enrolled in the classroom.
A lecture session is active.
Lecture materials have been generated and are available.</td></tr>
<tr><td>Post-conditions</td><td>The student's query is addressed with an appropriate response.</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The student signals to ask a question during the lecture.</td><td></td></tr>
<tr><td></td><td>2. The system acknowledges the student's signal to ask a question. (Stops script after finishing sentence)</td></tr>
<tr><td>3. The student submits their question either verbally by toggling their microphone or by typing it as text.</td><td></td></tr>
<tr><td></td><td>4. If the question is verbal, the system converts the voice input to text.</td></tr>
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<table>
<tr><td></td><td>The system processes the question to understand its context and intent.</td></tr>
<tr><td></td><td>5. The system generates an appropriate answer based on the lecture content and available resources.</td></tr>
<tr><td></td><td>6. The system delivers the response audibly to the student, ensuring clarity and relevance.</td></tr>
</table>
</td>
</tr>
<tr>
<td>Alternate Flows:</td>
<td>1. Clarification Needed:
○ Trigger: The system determines that the submitted question is unclear or ambiguous.
○ User Action: The system prompts the student to provide more details or rephrase the question.
○ System Response: Waits for the student to clarify and then processes the revised question accordingly.</td>
</tr>
<tr>
<td>Exceptions:</td>
<td>1. Unrecognized Question:
○ Trigger: The system fails to generate a response to the student’s question.
○ System Response: Informs the student that the question could not be understood and suggests rephrasing or trying again later.
○ User Action: The student rephrases the question.
2. System Downtime:
○ Trigger: The system is unavailable or experiencing technical difficulties during the lecture.
○ System Response: Notifies the student that the Q&A feature is currently unavailable and advises to ask questions once the system is back online.
○ User Action: The student notes down the question to ask later or waits until the system is operational again.</td>
</tr>
</table>

2.1.8. UC-07 Answer Pop-up Questions

<table>
<tr><td>Use Case Name:</td><td>Answer Pop-up Questions</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Student</td></tr>
<tr><td>Stakeholders:</td><td>Teacher: Wants to monitor student engagement to identify participation levels of students.
Student: Benefits from being actively engaged during lectures, which enhances learning.</td></tr>
<tr><td>Description:</td><td>Throughout the lecture, the system presents scheduled pop-up questions to assess student engagement. Students respond to these questions, and the system records their participation and accuracy to calculate engagement scores.</td></tr>
<tr><td>Pre-conditions:</td><td>The student is enrolled in the classroom.
A lecture session is active.
Lecture materials have been generated and are available.</td></tr>
<tr><td>Post-conditions</td><td>The student's responses to pop-up questions are recorded within the lecture.
Engagement scores are updated based on the student's participation and accuracy.</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The student is attending an active lecture session.</td><td></td></tr>
<tr><td></td><td>2. At predefined intervals, the system presents a pop-up engagement question to the student.</td></tr>
<tr><td>3. The student reads the pop-up question.</td><td></td></tr>
</table>
</td></tr>
</table>

	<table><tr><td>4. The student selects an answer or provides input in response to the question.</td><td></td></tr><tr><td></td><td>5. The system records the student's response, including the time taken to answer and the accuracy of the response.</td></tr><tr><td></td><td>6. The system updates the student's engagement score based on their participation and correctness.</td></tr><tr><td></td><td>7. The system logs the interaction for further analytics and reporting.</td></tr></table>
Alternate Flows:	1. **Skipping a Question**: ○ **Trigger**: The student chooses not to answer a pop-up question. ○ **User Action**: The student skips the question. ○ **System Response**: Records the skipped status and adjusts the engagement score accordingly.
Exceptions:	1. **No Response Within Time Limit**: ○ **Trigger**: The student does not respond to the pop-up question within the allocated time. ○ **System Response**: Marks the response as unanswered and updates the engagement score to reflect the lack of participation. ○ **User Action**: The student can attempt to answer the remaining questions during the lecture. 2. **System Error During Question Delivery**: ○ **Trigger**: The system encounters an error while presenting the pop-up question. ○ **System Response**: Notifies the student that the engagement question could not be delivered and continues the lecture. ○ **User Action**: The student continues to participate in the lecture without interruption. 3. **Network Issues**: ○ **Trigger**: The student loses internet connectivity while responding to a pop-up question.

	○ **System Response**: Alerts the student of the connectivity issue and records the response as incomplete. ○ **User Action**: The student regains connectivity and participates in remaining engagement activities once restored.

2.1.9. UC-08 Attempt Quiz

<table>
<tr><td>Use Case Name:</td><td>Attempt Quiz</td></tr>
<tr><td>Scope:</td><td>VirtuLectra</td></tr>
<tr><td>Level:</td><td>User Goal</td></tr>
<tr><td>Primary Actor:</td><td>Student</td></tr>
<tr><td>Stakeholders:</td><td>Student: Needs to complete quizzes within the allocated time to receive timely feedback.
Teacher: Seeks to assess student understanding and provide immediate feedback through automated grading.</td></tr>
<tr><td>Description:</td><td>A multiple-choice quiz is added to the student's quiz tab with a deadline. When the student opens the quiz, a timer starts to indicate the remaining time. If the quiz isn't submitted before time expires, the system automatically submits the current answers.</td></tr>
<tr><td>Pre-conditions:</td><td>The student is enrolled in the classroom.
A quiz has been generated and added to the student's quiz tab.
The quiz deadline is active and has not yet passed.</td></tr>
<tr><td>Post-conditions</td><td>The quiz is submitted, either manually by the student or automatically by the system.</td></tr>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The student Navigates to the Quiz tab.</td><td></td></tr>
<tr><td>2. Selects active Quiz.</td><td></td></tr>
<tr><td></td><td>3. The system displays the quiz questions and starts the timer indicating the remaining time.</td></tr>
<tr><td>4. The student answers the multiple-choice questions within the allocated time.</td><td></td></tr>
</table>
</td></tr>
</table>

<table>
<tr>
<td></td>
<td>
<table>
<tr><td>5. The student submits the quiz before the timer expires.</td><td></td></tr>
<tr><td></td><td>6. The system records the submitted answers and stops the timer.</td></tr>
</table>
</td>
</tr>
<tr>
<td>Alternate Flows:</td>
<td>1. Auto-Submission Due to Time Expiry:
○ Trigger: The student does not submit the quiz before the timer expires.
○ System Response: Automatically submits the current answers.
○ User Action: The student receives a notification that the quiz has been auto submitted.</td>
</tr>
<tr>
<td>Exceptions:</td>
<td>1. Quiz Access Denied:
○ Trigger: The student attempts to access the quiz after the deadline has passed.
○ System Response: Displays a message indicating that the quiz is no longer available for submission.
○ User Action: The student cannot attempt the quiz and may contact the teacher for further assistance.
2. Network Issues During Quiz:
○ Trigger: The student loses internet connectivity while taking the quiz.
○ System Response: Alerts the student of the connectivity issue and attempts to save progress locally.
○ User Action: The student regains connectivity and continues the quiz, with the system updating the timer accordingly.</td>
</tr>
</table>

2.1.10. UC-09 Handle Quiz Grades

Use Case Name:	Handle Quiz Grades
Scope:	VirtuLectra
Level:	User Goal
Primary Actor:	Student, Teacher
Stakeholders:	• **Teacher**: Wants to efficiently evaluate student performance and monitor progress through automated grading. • **Student**: Seeks timely feedback on quiz performance to understand their learning progress.
Description:	The system automatically grades submitted quizzes based on the correct answers and provides graded reports to the students. Teachers can view and edit quiz reports to ensure accuracy, while students can view their own quiz performance to track their learning progress.
Pre-conditions:	• Quizzes have been submitted by students.
Post-conditions	• Students have access to their graded quiz reports. • Teachers have access to the marks and can edit them if necessary.

<table>
<tr><td>Main Scenario:</td><td>
<table>
<tr><th>User Action</th><th>System Response</th></tr>
<tr><td>1. The student submits an attempted quiz.</td><td></td></tr>
<tr><td></td><td>2. The system checks the quiz based on the predetermined correct answers.</td></tr>
<tr><td></td><td>3. The system displays the student's quiz scores along with any provided feedback.</td></tr>
<tr><td>4. The student reviews their quiz performance to identify strengths and areas for improvement.</td><td></td></tr>
</table>
</td></tr>
<tr><td>Alternate flows:</td><td>1. Manual Grade Override:
○ Trigger: The teacher identifies an error in the automated grading.
○ User Action: The teacher selects the specific quiz score and adjusts it manually.
○ System Response: The system updates the score and logs the manual override.</td></tr>
<tr><td>Exceptions:</td><td>1. System Error While Viewing Reports:
○ Trigger: The system fails to load quiz reports.
○ System Response: Displays an error message and advises to try again later.
○ User Action: The user tries accessing the reports.</td></tr>
</table>

2.2. Functional Requirements

2.2.1. Real-Time Interaction Module

Conversational Voice Exchanges

A) Voice Input

2.2.1.1. Upon detecting voice input during the lecture, the system shall process the voice data and then convert it into text.

2.2.1.2. The system shall implement a time-out mechanism to stop listening for voice input after a defined period of inactivity.

2.2.1.3. If voice input is not detected or cannot be processed due to technical issues, the system shall notify the student accordingly.

2.2.1.4. The system shall provide visual feedback on the student interface when the microphone is activated to indicate that it is listening.

B) Voice Output

2.2.1.5. The system shall convert the lecture script into synthesized speech.

2.2.1.6. The system shall deliver the synthesized speech of the lecture script verbally to students.

2.2.1.7. During live lectures, the system shall convert all generated text-based responses to student queries into synthesized speech.

2.2.1.8. The system shall ensure that the conversion occurs in real time, with minimal delay in delivering verbal responses to students.

2.2.1.9. The system shall provide visual feedback when synthesized speech is being delivered.

2.2.1.10. The system shall ensure that all synthesized speech outputs are clear.

Live Q&A Sessions

A) Question Analysis and Response

2.2.1.11. When the student asks a question, the system shall detect and identify the main subject or topic of the question.

2.2.1.12. When a student asks a complex or multi-part question, the system shall break it down into separate, smaller components.

2.2.1.13. The system shall recognize key terms or phrases within the student’s question that relate to the ongoing lecture or previously covered material.

2.2.1.14. Based on the identified topics, the system shall generate an appropriate and contextually relevant answer to the student’s question using LLM.

B) Clarification and Follow-Up

2.2.1.15. If the LLM determines a question to be vague, the system shall prompt the student to provide clarification.

2.2.1.16. When the system cannot generate a valid response to a question, the system shall prompt the student to rephrase their question.

2.2.1.17. The system shall maintain a history of all questions asked during the lecture to ensure a continuous flow of conversation.

2.2.1.18. If the student's question pertains to the lecture notes, the system shall direct the student to the relevant section of the notes in the response.

2.2.2. Automated Content Creation Module

Lecture Script and Notes Generation

2.2.2.1. The system shall extract key points and summaries for the topics provided using the LLM.

A) Lecture Notes

2.2.2.2. The system shall structure the extracted content into organized lecture notes.

2.2.2.3. The system shall automatically create headings for major sections based on topic divisions.

2.2.2.4. The system shall use bullet points to list key points under each heading.

2.2.2.5. The system shall include numbering where appropriate to indicate a sequence of points or steps.

B) Lecture Script

2.2.2.6. Based on the extracted content, the system shall generate a script using LLM.

2.2.2.7. The system shall provide smooth transitions between different sections or topics in the script.

2.2.2.8. The system shall format the script in a step-by-step manner, ensuring clear guidance for delivering the lecture.

2.2.2.9. The system shall ensure the script follows a logical flow from introduction to conclusion.

2.2.2.10. The system shall adjust the length and content of the script based on the specified time duration of the lecture.

Tailored Quiz Generation

2.2.2.11. Based on the provided topics, the system shall create relevant multiple-choice questions (MCQs) using LLM.

2.2.2.12. The system shall ensure the generated quiz covers a balanced mix of topics, and key points for these topics are covered.

2.2.2.13. For each MCQ, the system shall generate at least three incorrect but plausible answers (distractors).

2.2.2.14. The system shall ensure distractors are conceptually similar to the correct answer but incorrect.

2.2.2.15. The system shall avoid overly obvious incorrect answers to maintain the quiz's effectiveness.

Automated Grading of Quizzes

2.2.2.16. After quiz submission, the system shall compare the student's answers with the correct answers stored in the system.

2.2.2.17. Based on the number of correct answers, the system shall calculate the total score of the quiz.

2.2.2.18. If necessary, the system shall allow teachers to manually override the automated grades of the quizzes.

2.2.3. Student Monitoring Module

Engagement Tracking

2.2.3.1. The system shall present pop-up questions to students periodically during live lectures.

2.2.3.2. The system shall allow the student to respond to the pop-up questions.

2.2.3.3. The system shall log the student's response to the pop-up questions.

2.2.3.4. The system shall calculate the student's engagement score based on how quickly and accurately they respond to the pop-up questions.

2.2.4. Web Portal Module

User Management

2.2.4.1. The system shall allow users (teachers and students) to register for an account by providing personal details such as name, email, and password.

2.2.4.2. When registering the user account, the system shall verify the provided email address to complete registration.

2.2.4.3. The system shall allow users to log in using their registered email and password.

2.2.4.4. The system shall provide users with the ability to reset their password via a password reset link sent to their registered email.

2.2.4.5. The system shall provide users with the ability to log in using OAuth services, such as Google.

2.2.4.6. The system shall allow users to update personal information on their profile.

2.2.4.7. The system shall allow users to delete their accounts permanently, with confirmation to prevent accidental deletion.

Teacher Dashboard

A) Classroom Management

2.2.4.8. The system shall allow the teacher to create a new classroom with a unique code for students to join.

2.2.4.9. The system shall allow the teacher to update the classroom name, description, and related information.

2.2.4.10. The system shall provide the teacher with the ability to upload course material for AI processing.

2.2.4.11. The system shall accept multiple formats for course materials, including PDF, and DOCX.

2.2.4.12. The system shall provide the teacher with the ability to specify the topics or chapters within the course material to focus on.

2.2.4.13. When uploading the course material, the system shall provide confirmation of successful upload or error messages for failed attempts.

2.2.4.14. The system shall provide the teacher with the ability to delete classrooms that are no longer required.

B) Quiz Management

2.2.4.15. The system shall give teachers the option to generate quizzes.

2.2.4.16. The system shall allow teachers to customize settings for quiz generation, such as the number of questions, topics to be covered, and time limit.

2.2.4.17. The system shall allow teachers to access, edit, and delete quizzes generated.

2.2.4.18. The system shall allow the teacher to access student records, including quiz scores and participation statistics.

C) Lecture Management

2.2.4.19. The system shall provide the teacher with the ability to schedule a lecture by selecting a date, time, and duration.

2.2.4.20. The system shall provide the teacher with the ability to set the lecture topics for the scheduled lecture.

2.2.4.21. The system shall notify all enrolled students about the scheduled lecture.

2.2.4.22. The system shall allow the teacher to reschedule or cancel lectures, with notifications sent to students accordingly.

2.2.4.23. The system shall allow the teacher to access lecture notes generated by the LLM.

Student Interface

A) Classroom Management

2.2.4.24. The system shall allow students to join classrooms by entering a valid classroom code provided by the teacher.

2.2.4.25. The system shall provide the student with the ability to leave the classroom they have joined.

B) Lecture Management

2.2.4.26. The system shall provide the student with the ability to attend scheduled lectures through a virtual classroom interface.

2.2.4.27. The system shall present lecture notes during the lecture.
2.2.4.28. The system shall allow the student with the ability to leave the lecture.
2.2.4.29. During the lecture, the system shall allow students to request permission to speak.
2.2.4.30. When a student requests to speak, the system shall complete the current sentence before prompting the student to proceed with their query.
2.2.4.31. Before the student proceeds with their query, the system shall offer the student to proceed with text or voice.
2.2.4.32. If the student chooses the text option, the system shall provide the student with the ability to type and submit their query.
2.2.4.33. If the student chooses the voice option, the system shall provide students with the ability to initiate voice input.
2.2.4.34. The system shall allow the student to view, and download lecture notes.

C) Quiz Management

2.2.4.35. The system shall provide the student with the ability to take a quiz for a specific period of time, after which the quiz shall be closed.
2.2.4.36. The system shall provide the student with feedback on their performance, including scores and correct answers.
2.2.4.37. The system shall provide the student with access to past quizzes and their corresponding scores and feedback.
2.2.4.38. The system shall notify students when a quiz is available.

2.3. Non-Functional Requirements

2.3.1. Reliability

2.3.1.1. The system shall provide a notification to administrators in case of downtime.
2.3.1.2. The system shall ensure that all critical data remains consistent and is not lost or corrupted during data transfers.
2.3.1.3. The system shall perform daily backups of all critical data.

2.3.2. Usability

2.3.2.1. The system shall provide a simple, intuitive user interface where labels, buttons, and options should be self-explanatory and clear.
2.3.2.2. The user interface shall be fully responsive, ensuring optimal usability across various devices and screen sizes.
2.3.2.3. The system shall acknowledge the user's action promptly with visual or auditory confirmation.
2.3.2.4. The system shall provide access to a help section that explains key features and guides users on performing common tasks.

2.3.2.5. The system shall communicate errors to users, providing actionable suggestions to resolve issues.

2.3.3. Performance

2.3.3.1. The system shall ensure that user inputs, such as voice or text, are processed with minimal latency.
2.3.3.2. The system shall ensure that any page within the teacher or student dashboard loads in less than 3 seconds under normal network conditions.
2.3.3.3. The system shall allow course materials of up to 100 MB in size to be uploaded.

2.3.4. Security

2.3.4.1. The system shall require users to create strong passwords that meet security standards.
2.3.4.2. The system shall encrypt all sensitive data using industry-standard encryption protocols.
2.3.4.3. The system shall enforce role-based access control (RBAC) to ensure that only authorized users can access certain features.
2.3.4.4. The system shall implement automatic account lockout after multiple failed login attempts to prevent brute-force attacks.
2.3.4.5. The system shall implement protection against common security threats, including SQL injection, cross-site scripting (XSS), and cross-site request forgery (CSRF).

2.4. Domain Model

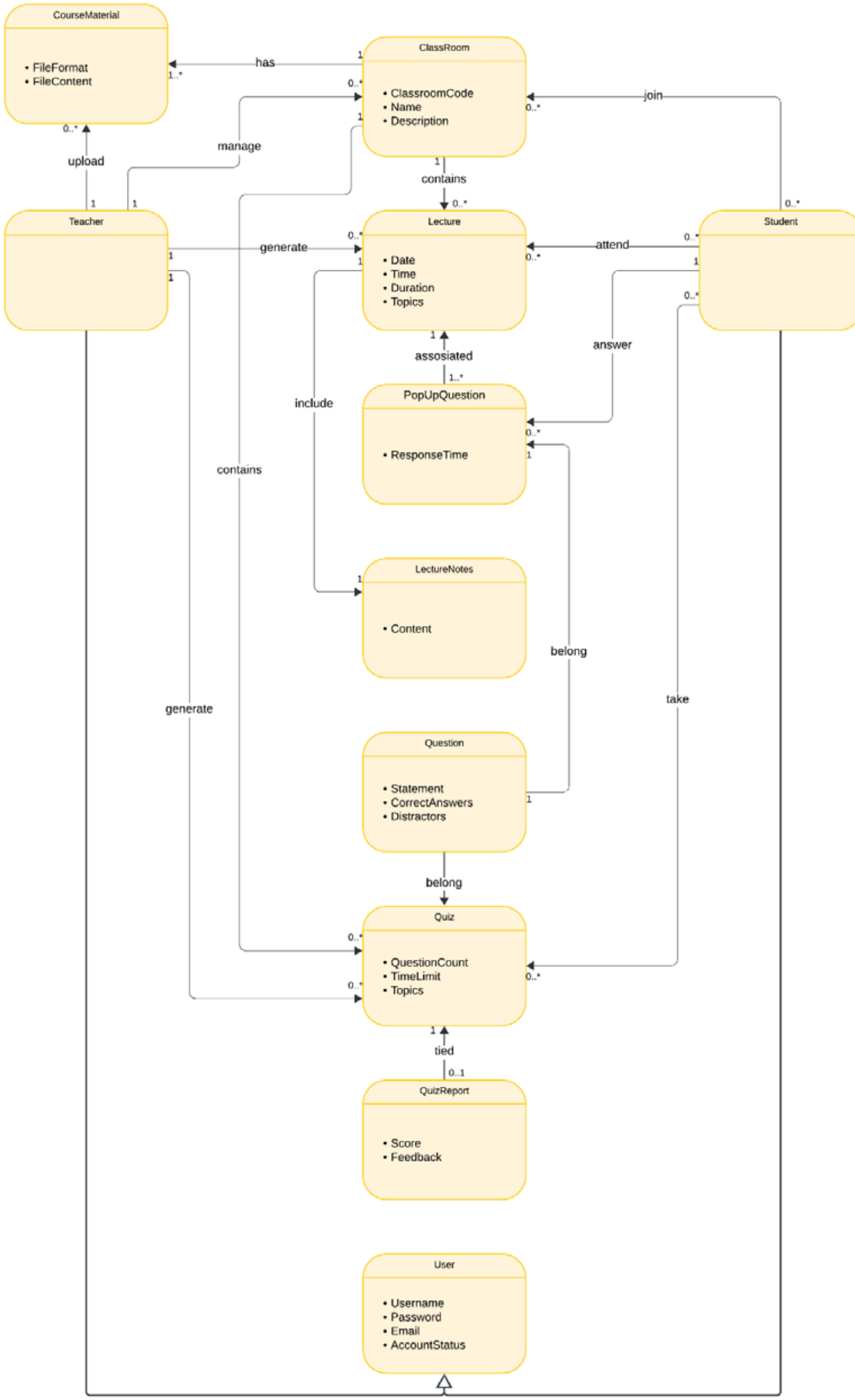

Figure 2 Domain Model

Chapter 3

System Overview

3.1. Architecture Design

Description:

The system uses a microservices architecture, where different services are responsible for specific tasks like user management, engagement tracking, grading, quiz generation, and lecture notes creation. These services communicate through an API Gateway, which directs client requests to the appropriate service. Core functions like authentication, logging, and notifications are handled separately in a common services layer, making it easier to maintain and scale the system. The architecture also integrates with external systems through third-party APIs and OAuth, allowing for additional functionality. Data is managed using MongoDB and VectorDB, ensuring efficient storage and processing. Each service operates independently, making the system flexible and scalable.

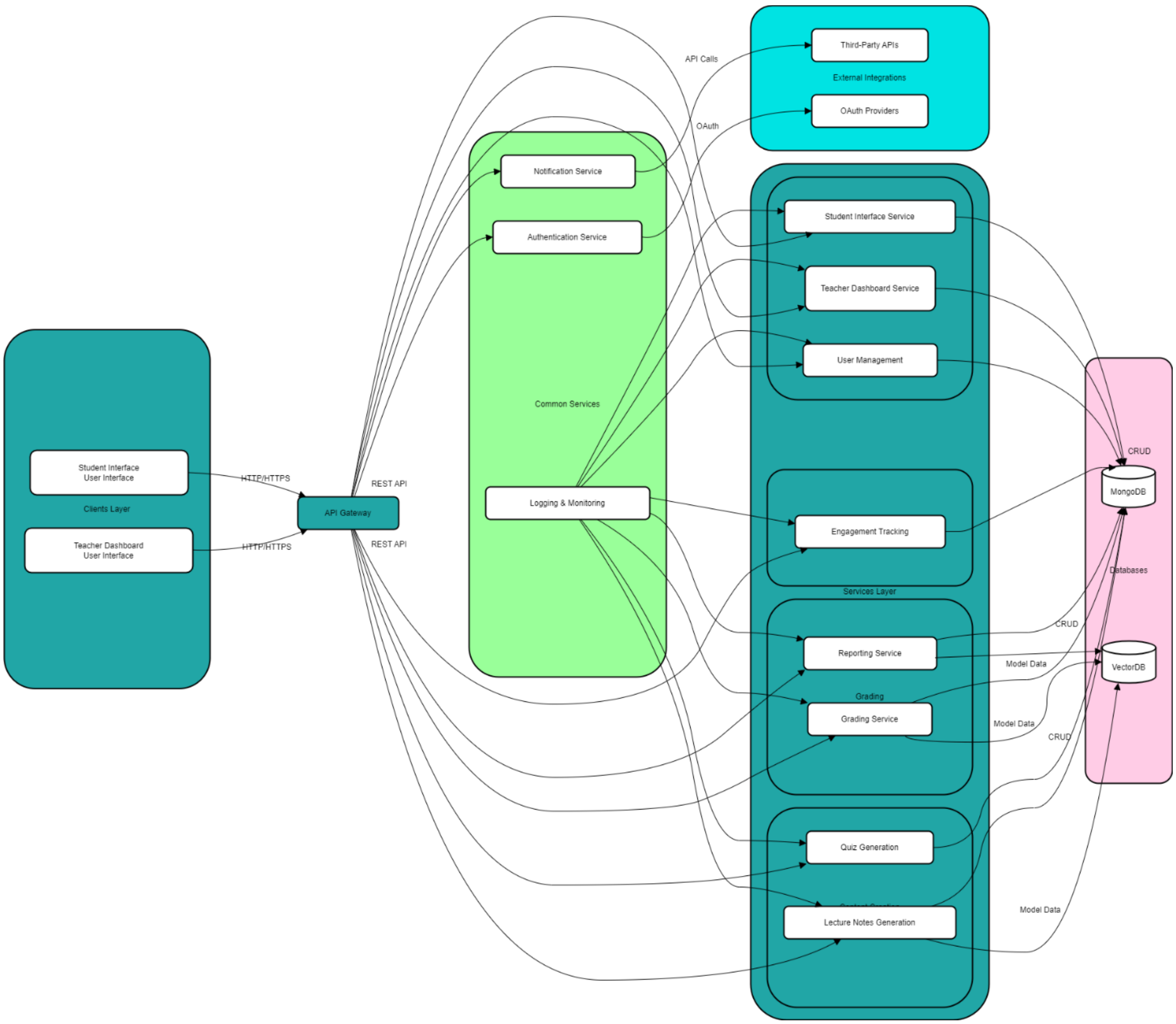

Figure 3 Architecture Diagram

3.2. Design Models

Activity Diagrams

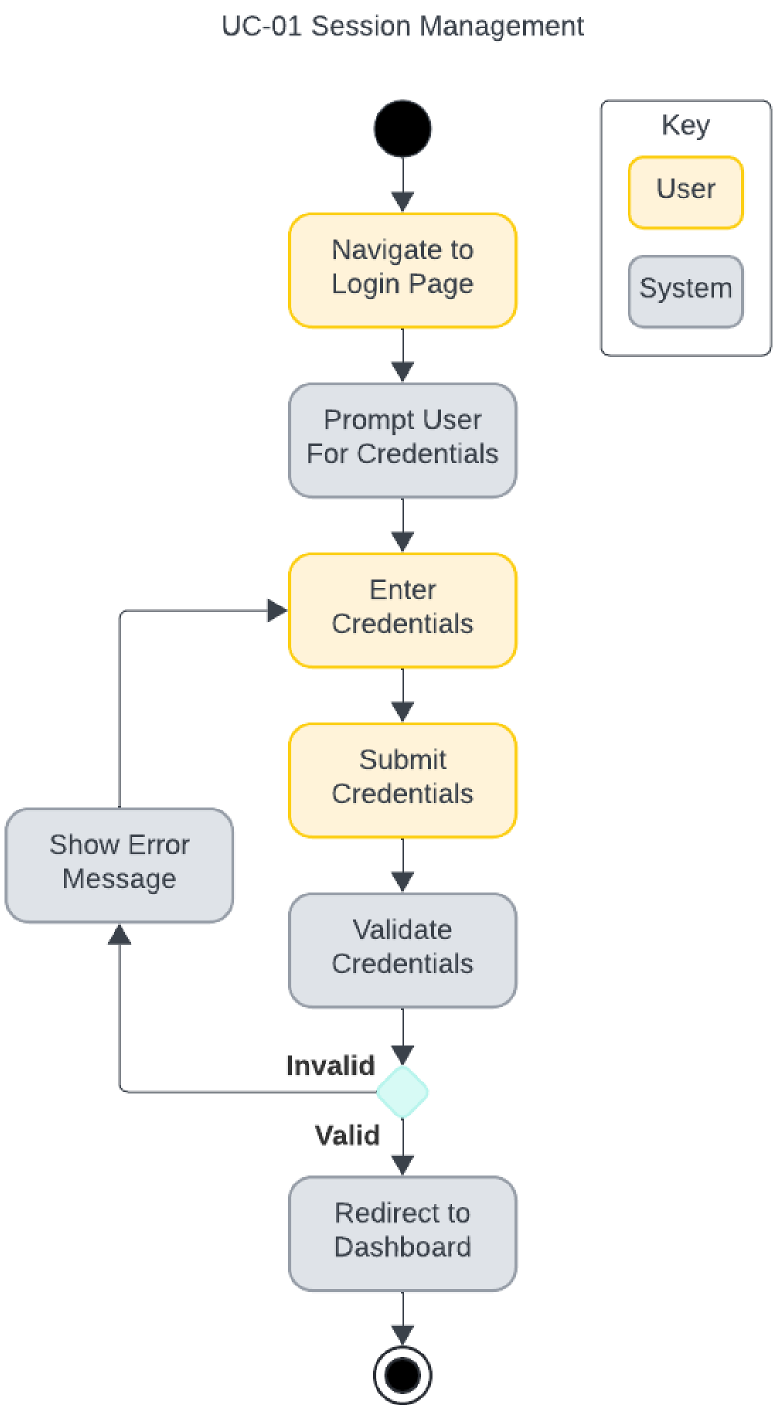

Figure 4 Act-01 Session Management

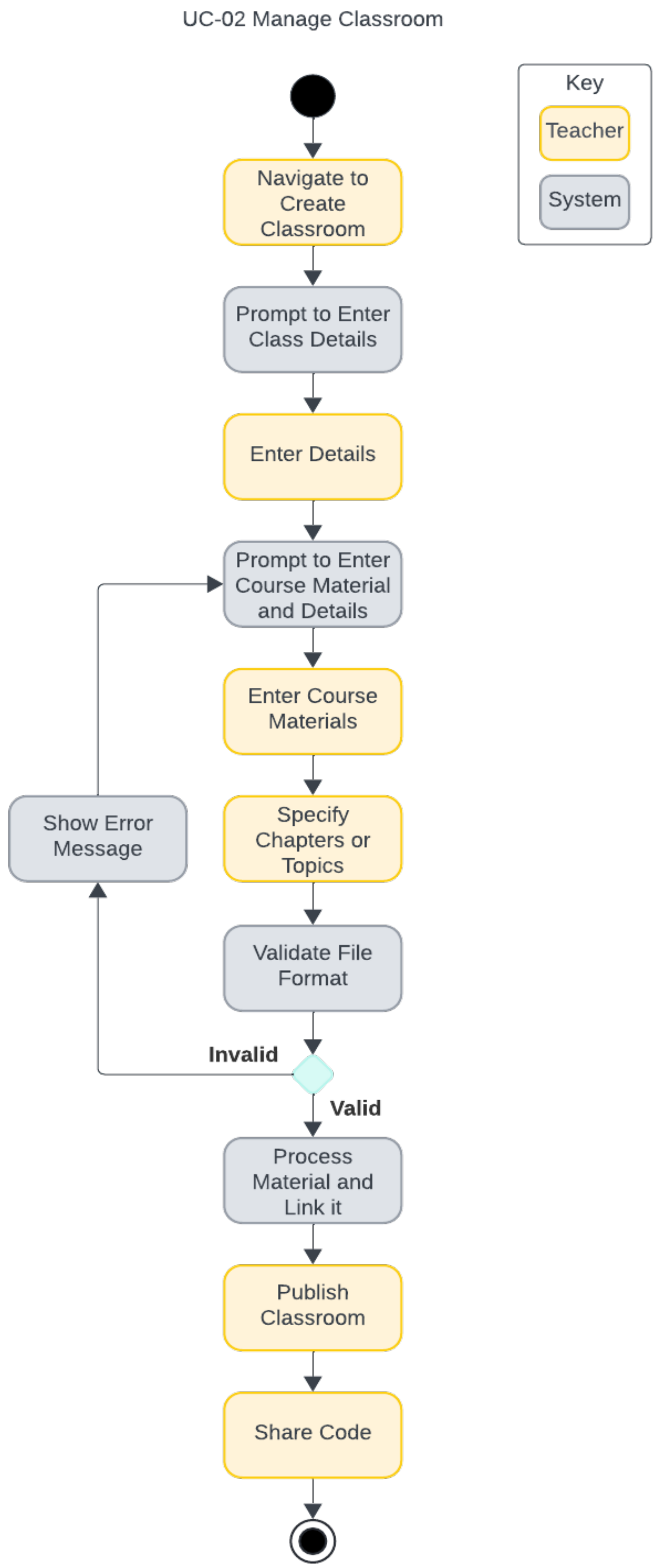

Figure 5 Act-02 Manage Classroom

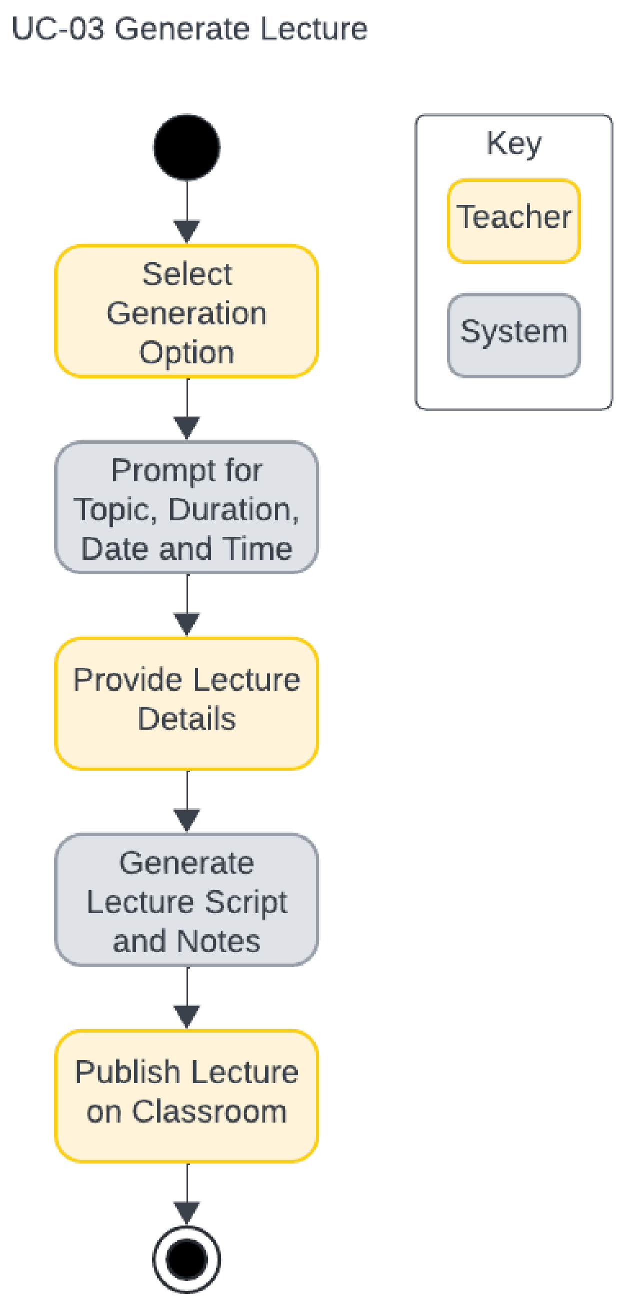

Figure 6 Act-03 Generate Lecture

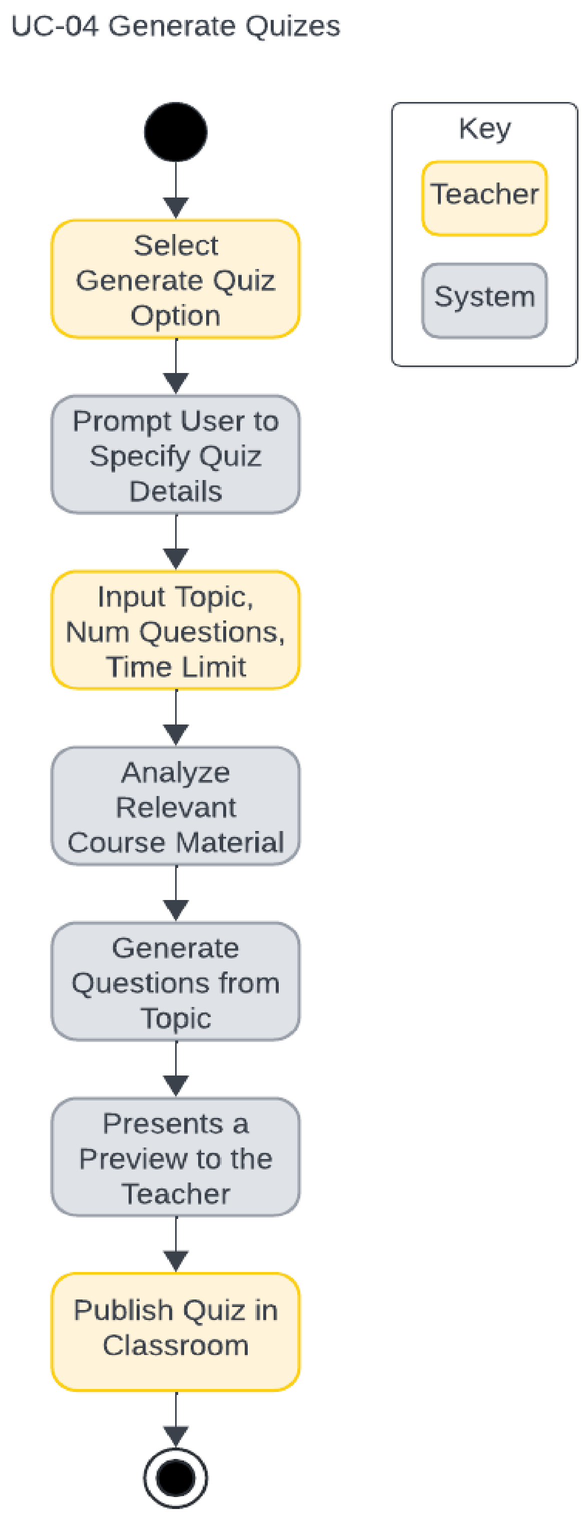

Figure 7 Act-04 Generate Quizzes

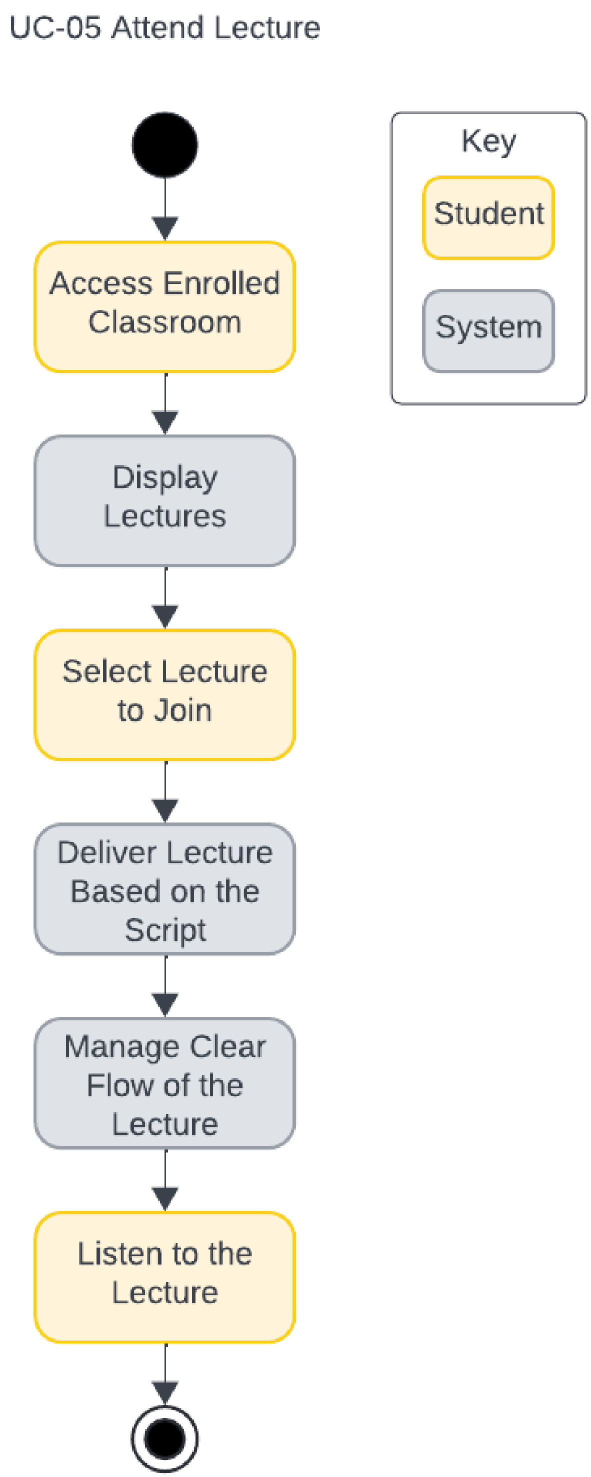

Figure 8 Act-05 Attend Lecture

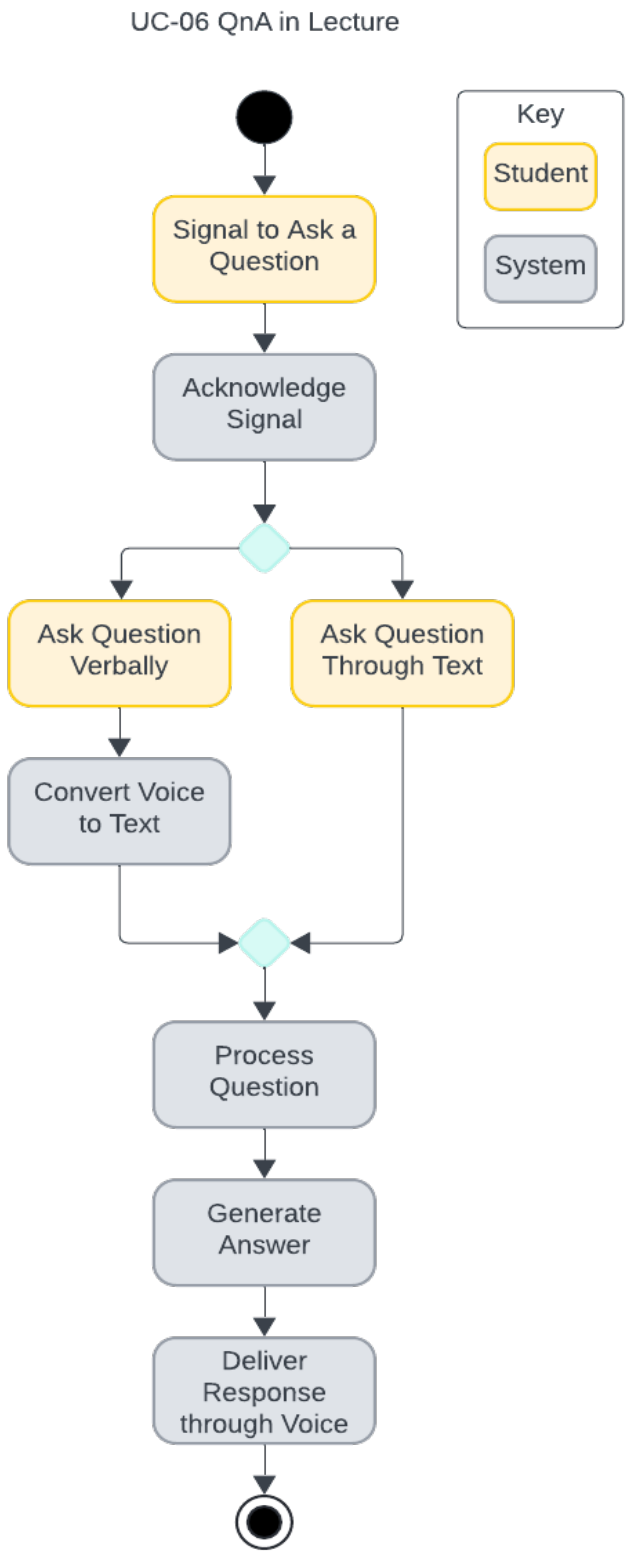

Figure 9 ACt-06 Q&A in Lecture

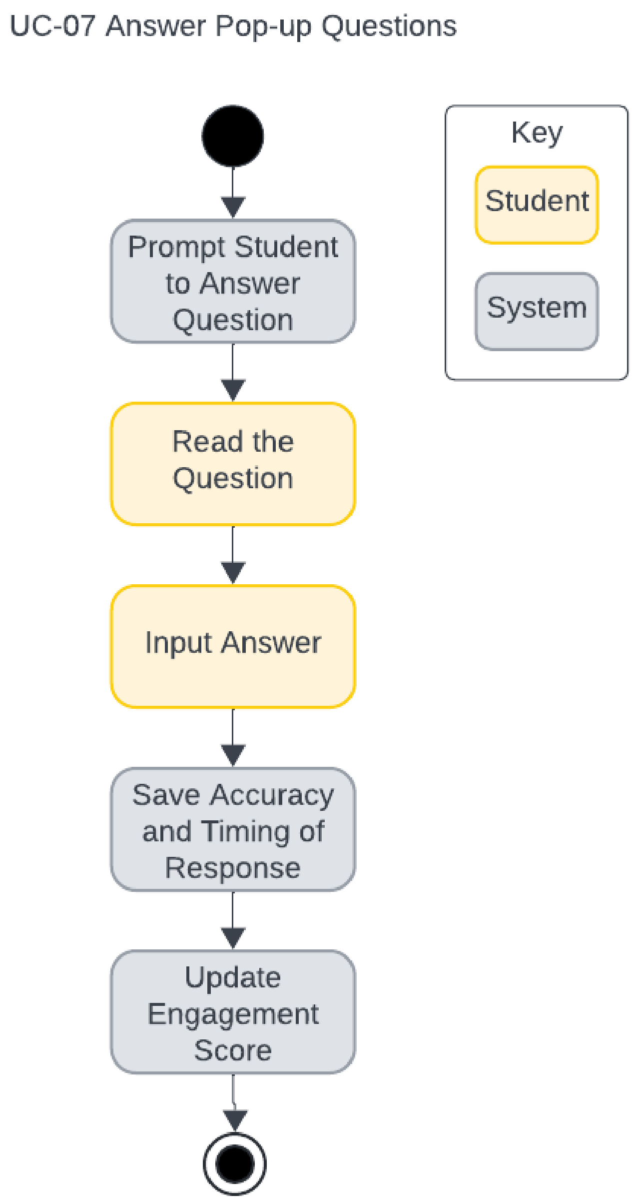

Figure 10 Act-07 Answer Popup Questions

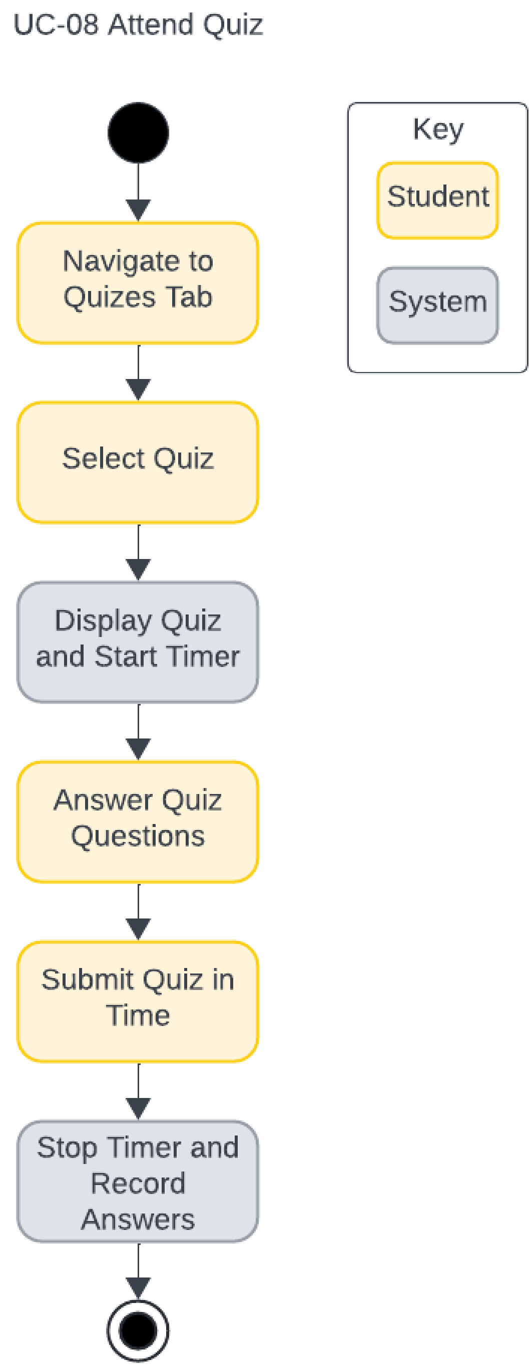

Figure 11 Act-08 Attend Quiz

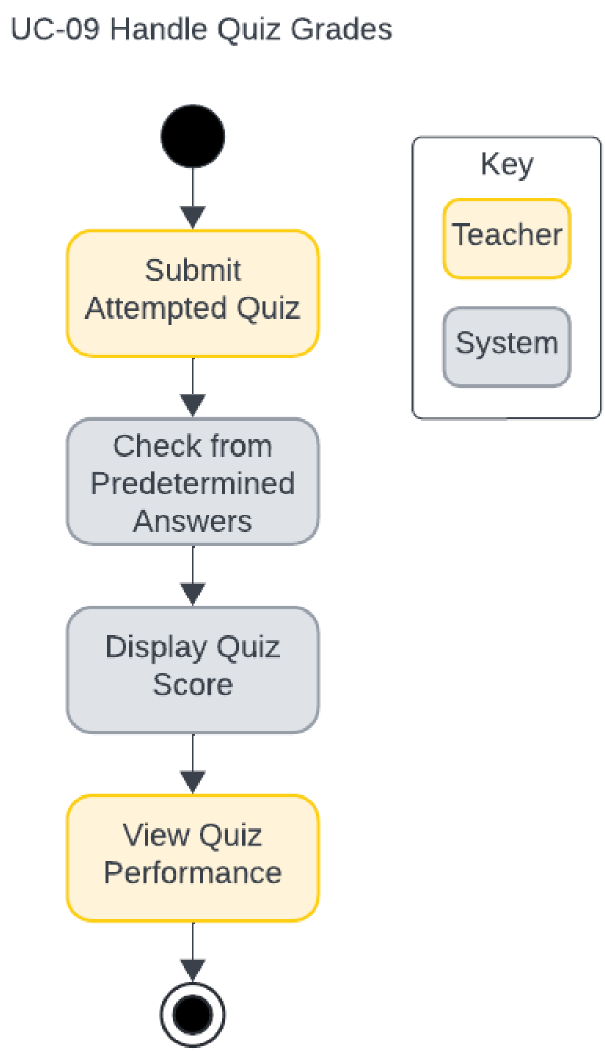

Figure 12 Act-09 Handle Quiz Grades

System Sequence Diagrams

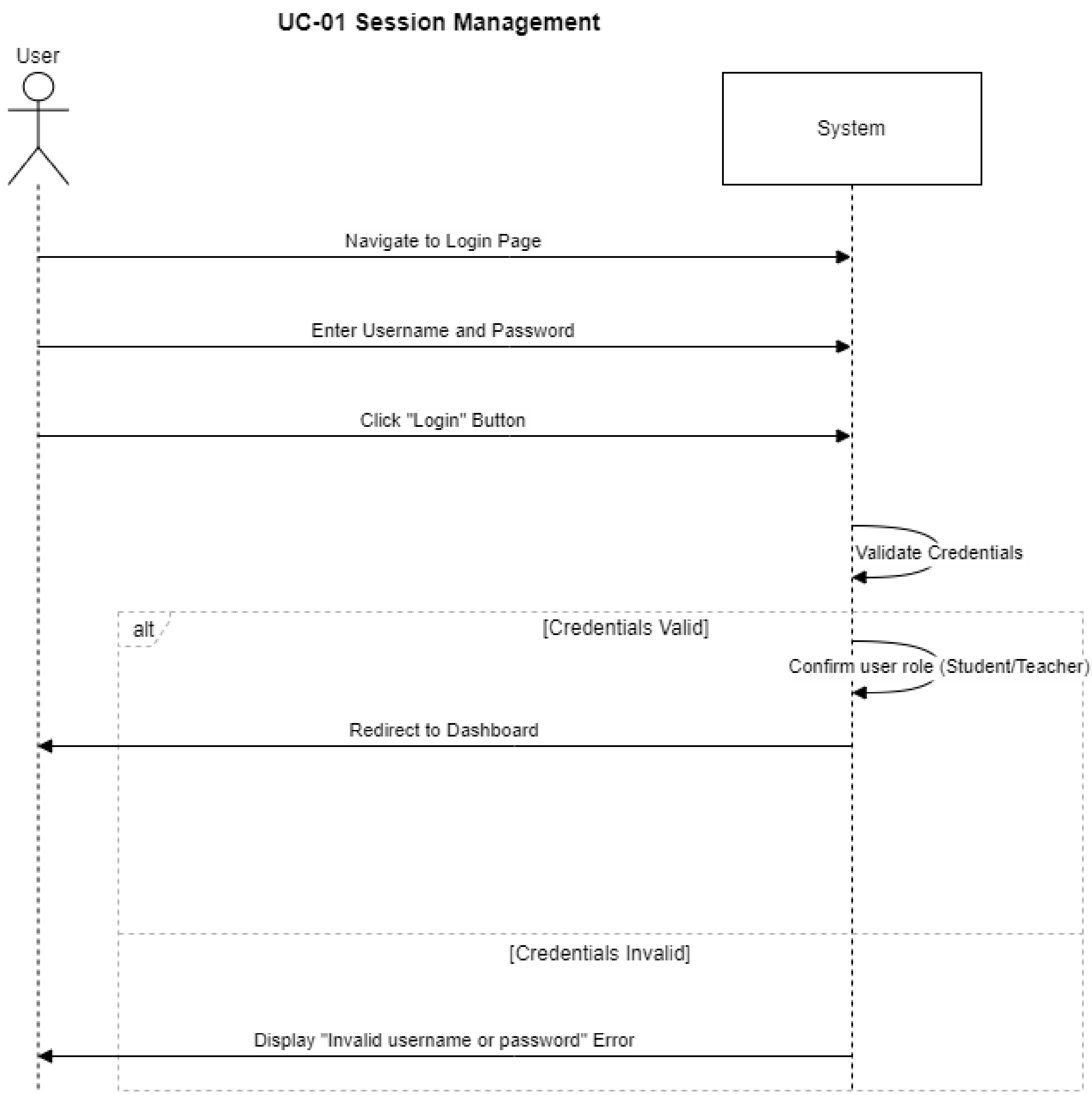

Figure 13 Seq-01 Session Management

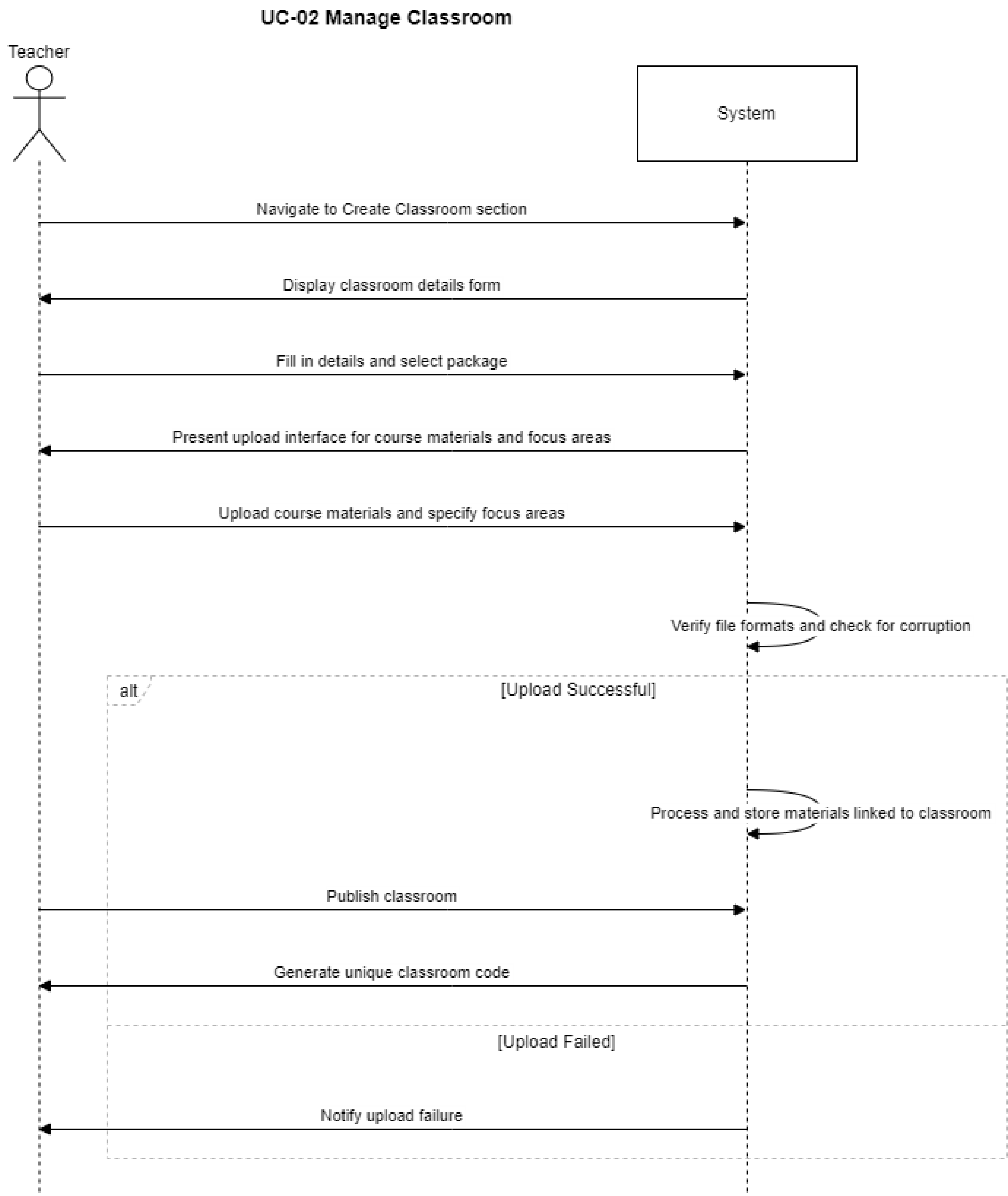

Figure 14 Seq-02 Manage Classroom

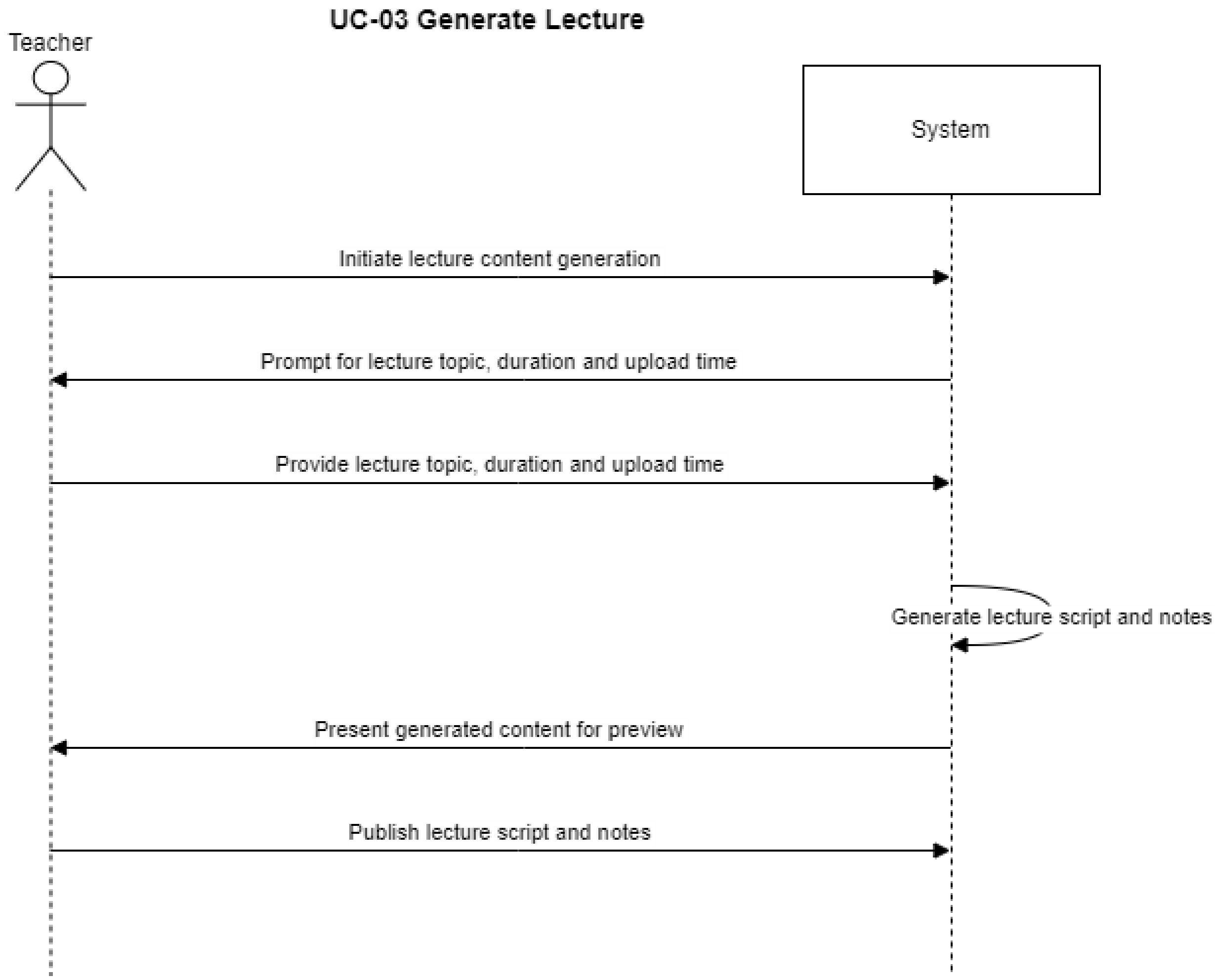

Figure 15 Seq-03 Generate Lecture

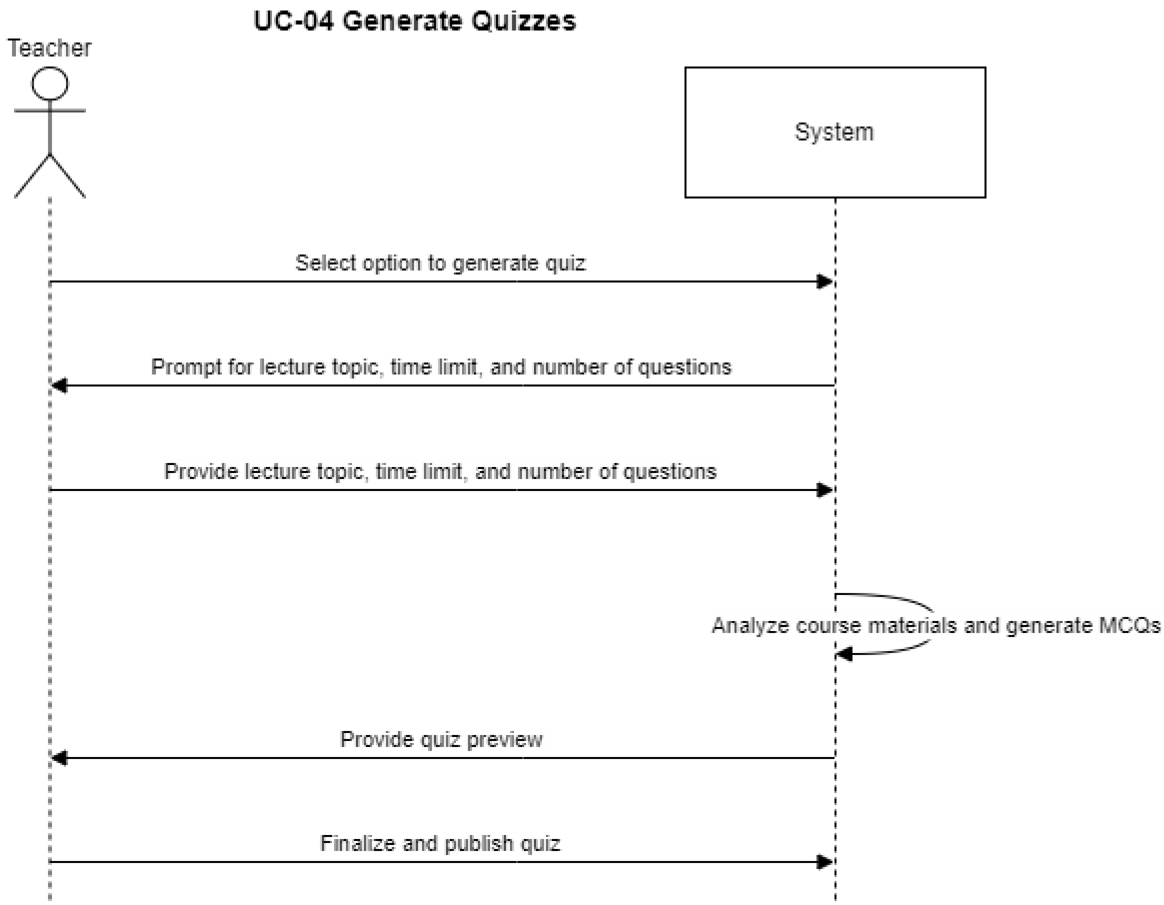

Figure 16 Seq-04 Generate Quizzes

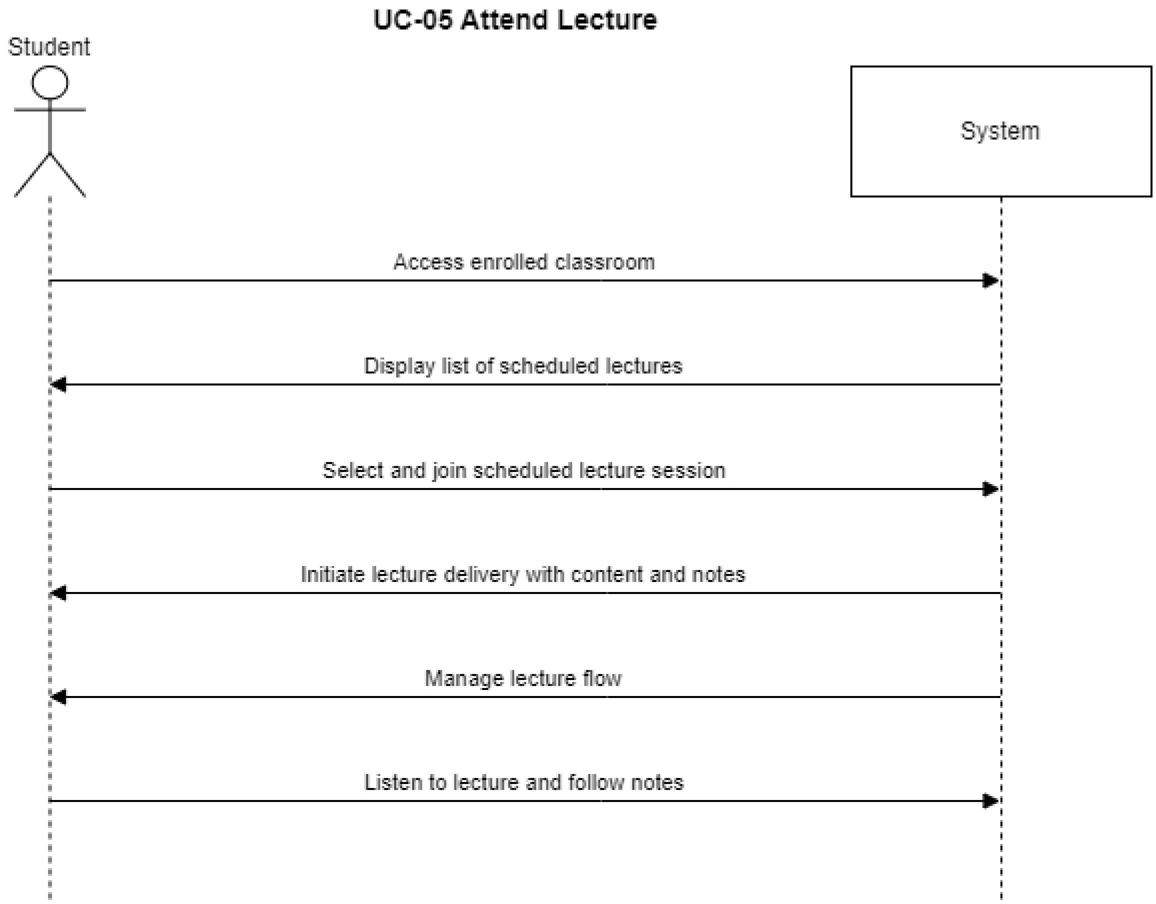

Figure 17 Seq-05 Attend Lecture

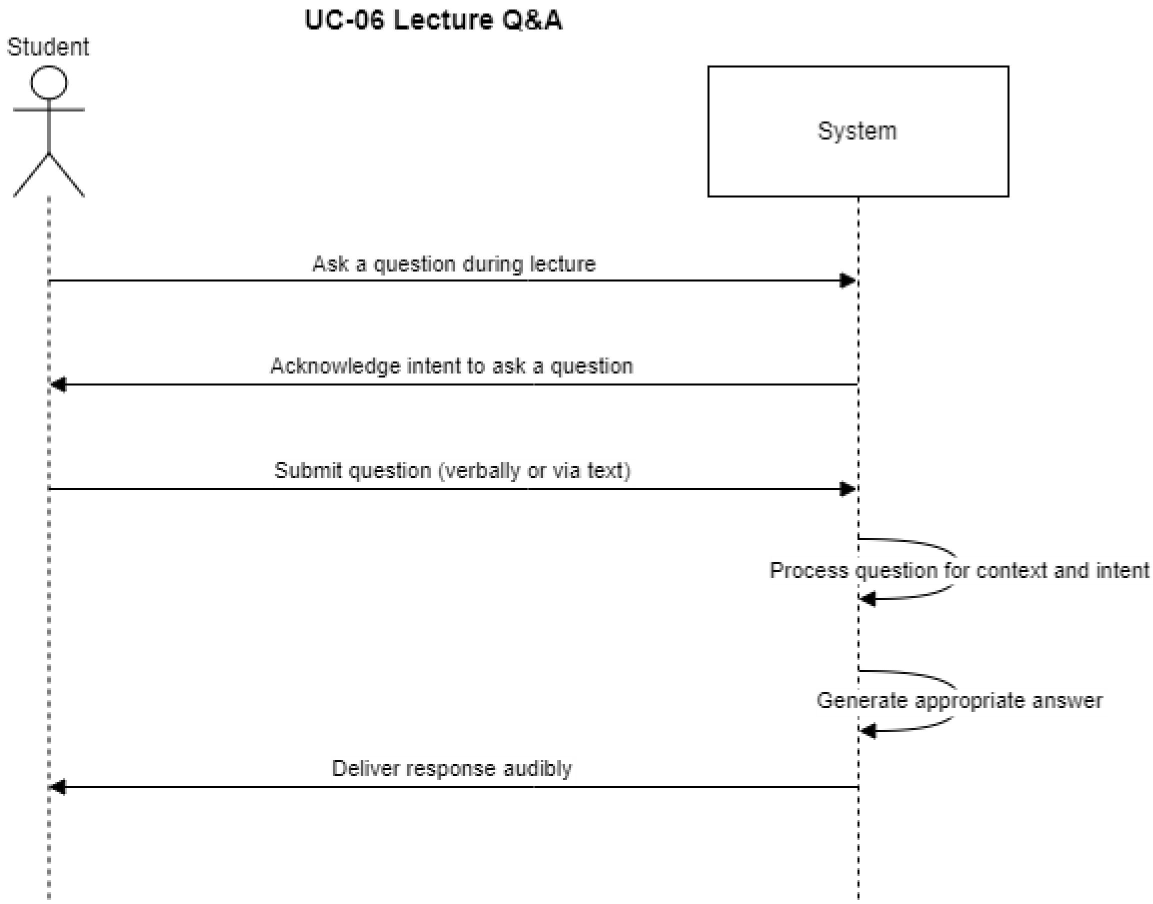

Figure 18 Seq-06 Lecture Q&A

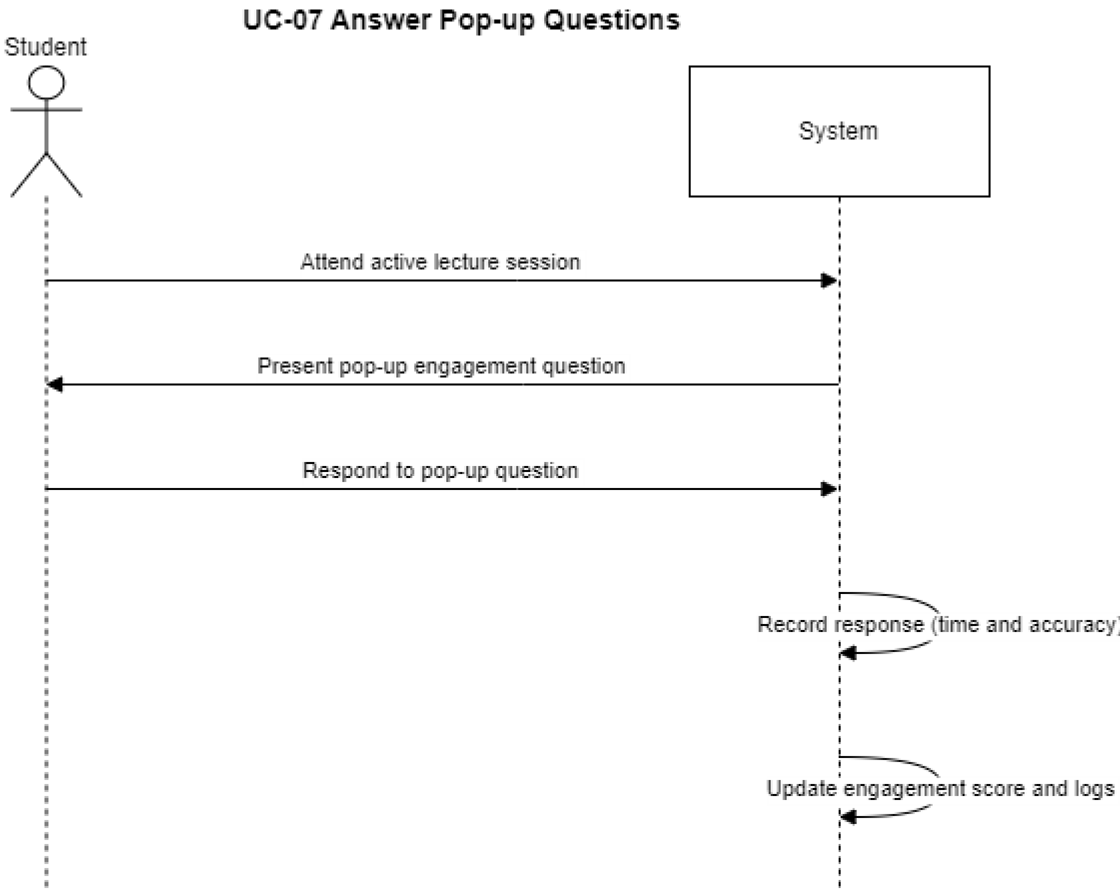

Figure 19 Seq07 - Answer Pop-up Questions

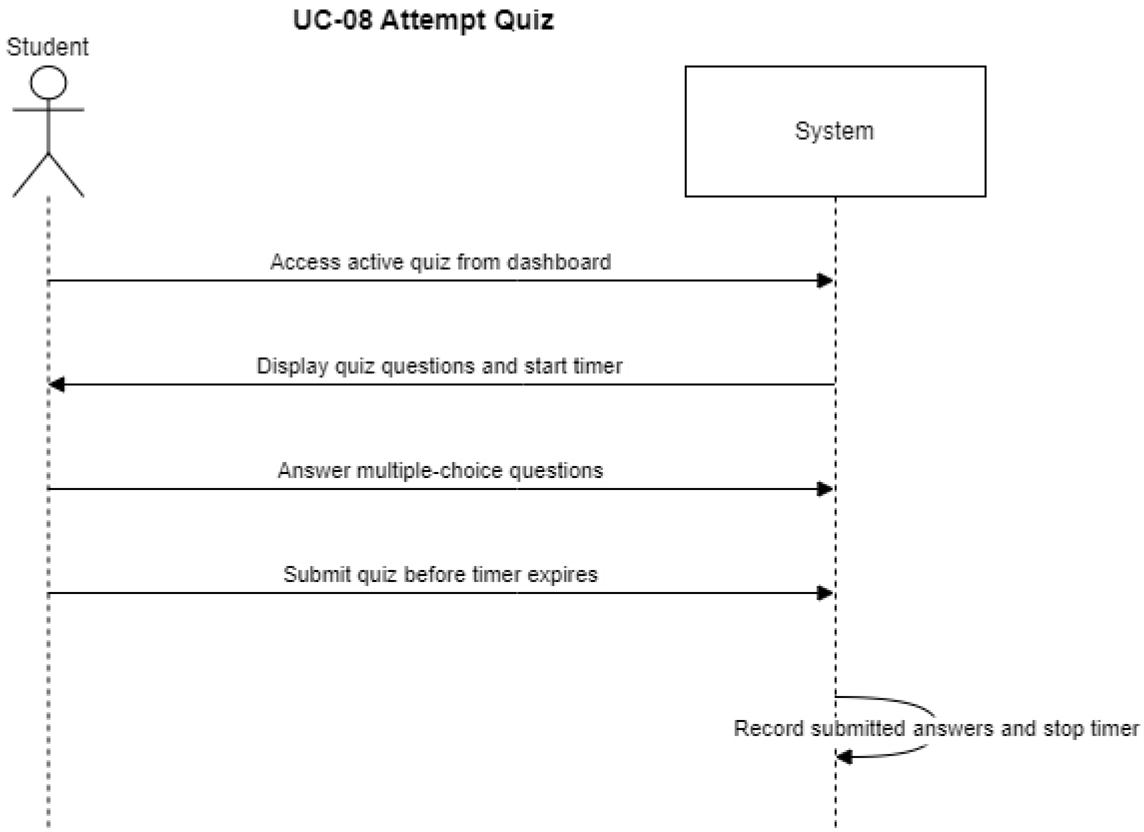

Figure 20 Seq-08 Attempt Quiz

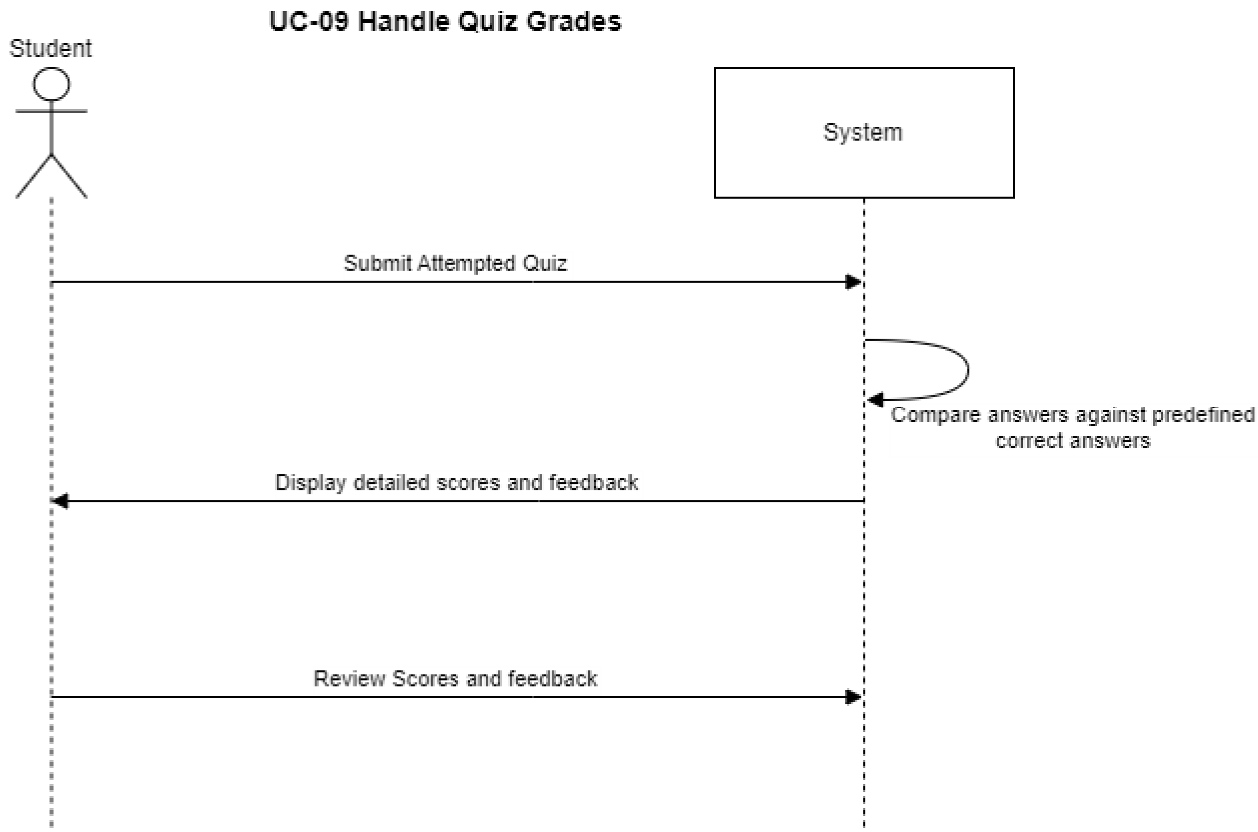

Figure 21 Seq-09 Handle Quiz Grades

Data Flow Diagram

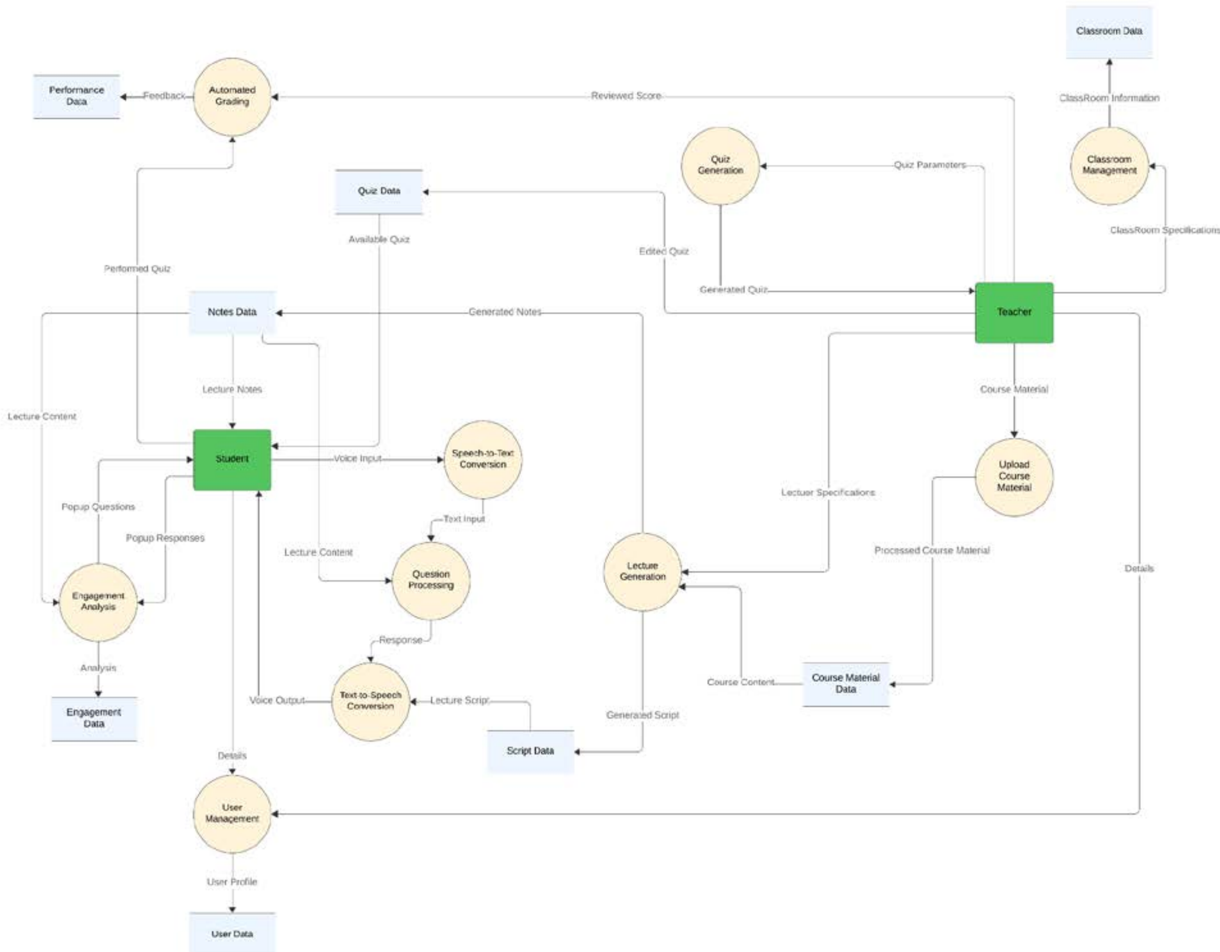

Figure 22 Data Flow Diagram

3.3. Data Design

Description:
The information domain is transformed into data structures that reflect the key entities and relationships in an educational platform, such as users (teachers and students), classrooms, lectures, quizzes, and course materials. Each entity is represented as a distinct object with unique identifiers (UUIDs), ensuring that they can be efficiently stored and retrieved in a database. Relationships between entities, such as teachers managing classrooms or students attending lectures, are represented using references between these objects, preserving the one-to-many and many-to-many relationships inherent in the domain.

Complex data like course materials and lecture notes are stored as vectors (arrays of numbers), making them suitable for use in AI-driven applications, such as machine learning models for content recommendation or similarity analysis. This vectorization allows for sophisticated processing and analysis of course content.

JSON Schema

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          "lectureNotes": { "$ref": "#/properties/LectureNotes" }
        },
        "required": ["id", "date", "time", "duration", "topics", "teacherId"]
      },
      "LectureNotes": {
        "type": "object",
        "properties": {
          "id": { "type": "string", "format": "uuid" },
          "content": {
            "type": "array",
            "items": { "type": "number" },
            "description": "Lecture notes in vector form (e.g., embeddings)"
          },
          "lectureId": {
            "type": "string",
            "format": "uuid",
            "description": "References the Lecture this note belongs to"
          }
        },
        "required": ["id", "content", "lectureId"]
      },
      "Quiz": {
        "type": "object",
        "properties": {
          "id": { "type": "string", "format": "uuid" },
          "questionCount": { "type": "number" },
          "timeLimit": { "type": "number", "description": "Time limit in minutes" },
          "topics": { "type": "array", "items": { "type": "string" } },
          "teacherId": {
            "type": "string",
            "format": "uuid",
            "description": "References a Teacher (User ID with Teacher role)"
          },
          "questions": {
            "type": "array",
            "items": { "$ref": "#/properties/Question" }
          },
          "quizReport": { "$ref": "#/properties/QuizReport" }
        },
        "required": ["id", "questionCount", "timeLimit", "topics", "teacherId"]
      },
      "Question": {
        "type": "object",
        "properties": {
          "id": { "type": "string", "format": "uuid" },
          "statement": { "type": "string" },
          "correctAnswers": {
            "type": "array",
            "items": { "type": "string" }
          },
          "distractors": {
            "type": "array",
            "items": { "type": "string" }
          },
          "quizId": {
            "type": "string",
            "format": "uuid",
            "description": "References the Quiz this question belongs to"
          },
          "popupQuestionId": {
            "type": "string",
            "format": "uuid",
            "description": "References the PopUpQuestion if applicable"
          }
        },
        "required": ["id", "statement", "correctAnswers", "distractors"]
      },
      "QuizReport": {
        "type": "object",
```

Figure 23 JSON Schema

```json
        "QuizReport": {
          "type": "object",
          "properties": {
            "id": { "type": "string", "format": "uuid" },
            "score": { "type": "number" },
            "feedback": { "type": "string" },
            "quizId": {
              "type": "string",
              "format": "uuid",
              "description": "References the Quiz this report is generated for"
            }
          },
          "required": ["id", "score", "feedback", "quizId"]
        },
        "PopUpQuestion": {
          "type": "object",
          "properties": {
            "id": { "type": "string", "format": "uuid" },
            "responseTime": { "type": "number" },
            "lectureId": {
              "type": "string",
              "format": "uuid",
              "description": "References the Lecture this popup question belongs to"
            },
            "studentAnswers": {
              "type": "array",
              "items": { "type": "string" }
            }
          },
          "required": ["id", "responseTime", "lectureId"]
        }
      },
      "required": ["User", "ClassRoom", "CourseMaterial", "Lecture", "LectureNotes", "Quiz", "Question", "QuizReport", "PopUpQuestion"]
    }
```